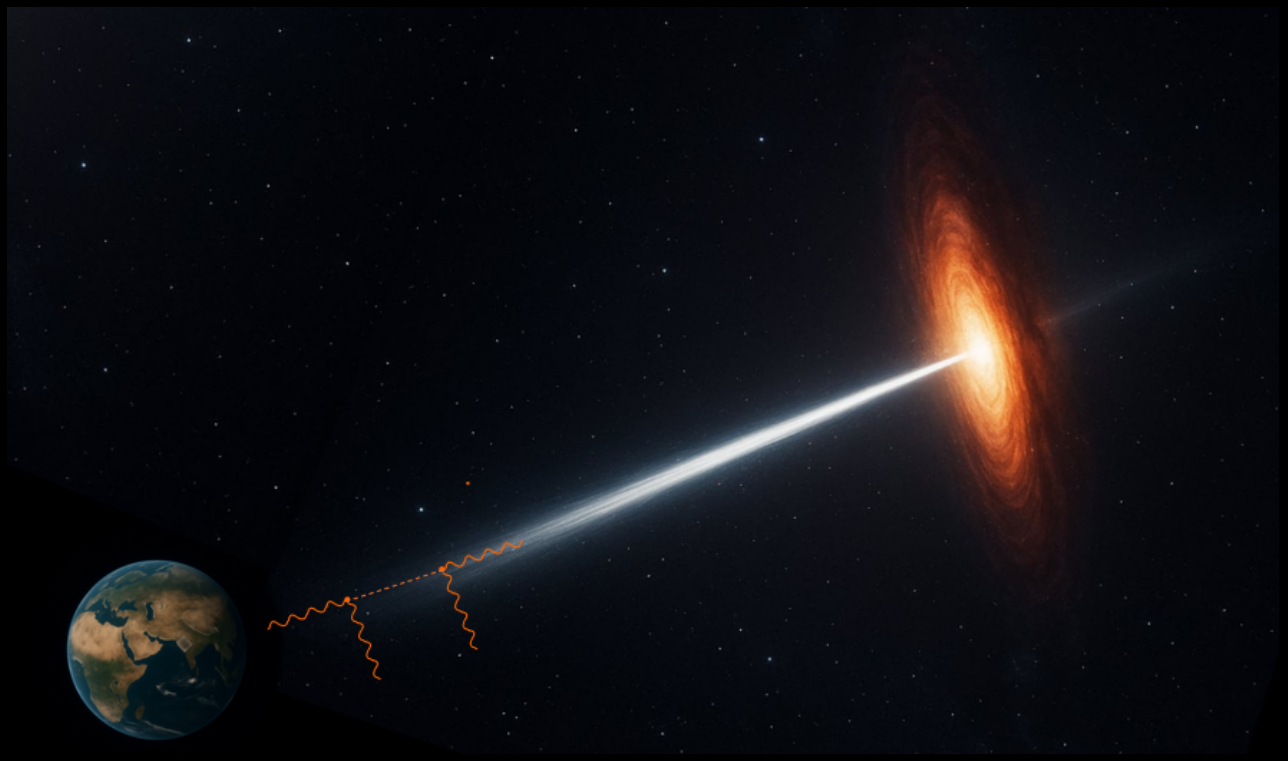
University of Southern Denmark

SPFY801

Master Thesis

# Do we need axion-like-particles to explain the very high photon energies from GRB221009A?




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## Abstract

In 2022, GRB221009A was detected by the LHAASO collaboration as the brightest gamma ray burst (GRB) ever recorded with photon energies that extended up to 18TeV. Combined with the redshift of z=0.151 of the GRB, it has been questioned if these observations can be reconciled with standard physics, since such high energy photons suffer from absorption on the extragalactic background light (EBL). A possible explanation could be the oscillation between gamma rays and hypothetical axion-like particles (ALPs), which would effectively lower the absorption. Such ALPs are predicted in numerous theories beyond the Standard model and are a viable candidate for cold dark matter.
Here, we analyze LHAASO data from GRB221009A using a maximum likelihood approach with a model of the observed spectra that incorporates both the emitted spectra and the photon survival probability. The photon survival probability was calculated using various EBL models and considering both scenarios with and without photon-ALP oscillations. Confronting the spectral models with the LHAASO data, both with and without photon-ALP oscillations, suggests that a model incorporating photon-ALP oscillations is generally not preferred across the ALP parameter space. However, a few combinations of ALP mass and photon-ALP coupling do favor a model that includes photon-ALP oscillations albeit at low significance. Therefore, we conclude that ALPs are generally not required to explain the LHAASO observations of GRB221009A.



## Resumé

I 2022 blev GRB221009A detekteret af LHAASO-samarbejdet som det kraftigste gamma strålings udbrud (GRB) nogensinde registreret med fotonenergier, der strakte sig op til 18TeV. Kombineret med en rødforskydningen på z=0.151 for GRB'en, har det været diskuteret, om disse observationer kan forenes med standardfysikken, da så høje energifotoner undergår af absorption på den ekstragalaktiske bagrundsstråling (EBL). En mulig forklaring kunne være oscillation mellem gamma stråler og hypotetiske axion-lignende partikler (ALPs), som efektivt vil reducere absorbtionen. Sådanne ALPs forudsiges i flere teorier, der går ud over standardmodellen, og de er en plausibel kandidat for kold mørk materie. Her analyserer vi LHAASO-data fra GRB221009A ved hjælp af en maksimum likelihood metode med en model for de observerede spektra, som inkluderer både det udsendte spektrum og fotonernes overlevelsessandsynligheden. Fotonernes overlevelsessandsynligheden blev beregnet med forskellige EBL modeller under både scenarier med og uden foton-ALP oscillationer. Sammenligning af de spektrale modeller med LHAASO data, både med og uden foton-ALP oscillationer, antyder generelt, at en model med foton-ALP oscillationer ikke foretrækkes i det meste af ALP parameterspektret. Dog favoriserer enkelte kombinationer af ALP-masse og foton-ALP kobling en model med foton-ALP oscillationer, omend med lav signifikans. Vi konkluderer derfor, at ALPs generelt ikke er nødvendige for at forklare LHAASO observationerne af GRB221009A.

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# abbreviations

**ALP** Axion-Like Particle. 6, 24–30

**EBL** Extragalactic Background Light. 6, 31–34

**GRB** Gamma-Ray Burst. 6, 31, 32

**GRBs** Gamma-Ray Bursts. 6

**SED** Spectral Energy Distribution. 31–33, 42

**VHE** Very High Energy. 6, 31–34

# Chapter 1

# Introduction

In the 1960s, the United States launched the Vela satellites to monitor nuclear detonations in compliance with the 1963 Nuclear Test Ban Treaty. Unexpectedly, the satellites recorded intense flashes of gamma radiation that bore no resemblance to known nuclear signatures. These mysterious signals were later identified as the first Gamma-Ray Bursts (GRBs) [52]. Since their discovery, Gamma-Ray Burst (GRB)s have become a central topic in high-energy astrophysics.

GRBs are among the most powerful and cataclysmic explosions in the universe - releasing just as much energy in a few seconds as the Sun does in its entire lifetime [56]. They are characterized by the emission of high-energy photons ranging from keV to TeV energies. The detection of Very High Energy (VHE) photons is a relatively recent discovery [4]. However, when VHE photons propagating through space are expected to interact with the Extragalactic Background Light (EBL), producing electron-positron pairs. This interaction leads to attenuation of the photon flux, making the detection of GRB photons above $\sim 10$ TeV highly improbable [29].

In 2022, the brightest GRB ever recorded, GRB221009A, challenged this expectation. It produced gamma rays with energies up to 18 TeV [45], making it the most energetic GRB observed to date. Given its redshift of $z = 0.151$ [61], these VHE photons should have undergone significant attenuation due to interactions with the EBL. Their detection suggests otherwise, leaving the origin of the observed high-energy emission an open and intriguing question.

One compelling solution involves photons oscillating into Axion-Like Particle (ALP)s. ALPs are hypothetical, light pseudoscalar particles that emerge in several extensions of the Standard Model of particle physics, including string theory [46]. In the presence of external magnetic fields, photons can convert into ALPs and vice versa [76]. Because ALPs interact very weakly with matter and radiation, they are not affected by EBL attenuation. When they reconvert into photons close to Earth, the observed spectrum can retain a higher gamma-ray flux than would otherwise be expected [18, 29].

This thesis investigates whether the inclusion of photon-ALP oscillations in models of the observed spectra can account for the VHE emission from GRBs and check whether ALPs are required at all to explain the high photon energies of GRB221009A.

# Chapter 2

# Gamma-ray bursts

One of the most energetic events in the Universe is a GRB. GRBs are short-lived, intense bursts of gamma-rays that originate from either the core collapse of a massive star [90] or a merger of compact binaries, such as neutron stars [30, 69]. The resulting central engine is typically a newly formed black hole or a neutron star .
GRBs exhibit highly variable and complex light curves, which are generally divided into two distinct phases: the bright prompt emission and the longer-lasting afterglow. The prompt emission is characterized by rapid variability and often consists of multiple spikes. In contrast, the afterflow displays a smoother and exponentially decaying emission [56].

## 2.1 Classification, Progenitor and host galaxy

Classification of GRBs has proven difficult due to their highly variable light curves. These light curves depend on the energy range and sensitivities of the detectors and may exhibit many erratic spikes, a single spike, a fast-rise-exponential-decay (FRED) profile, or appear smooth [56]. Their duration can also vary significantly, ranging from miliseconds to several minutes, see Figure 2.2. Despite this diversity, GRBs are commonly classified based on their duration. The quantity used is their $T_{90}$ time interval, defined as the time in which $5\% - 95\%$ of the total energy release is emitted [54, 92].

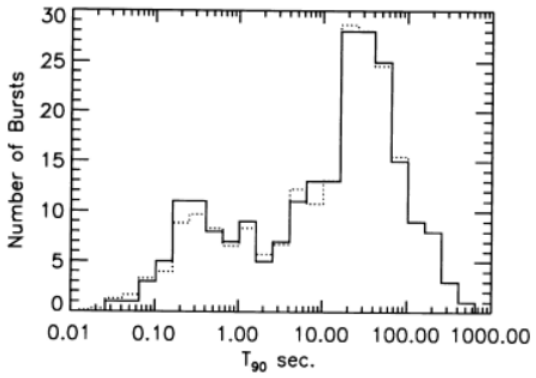
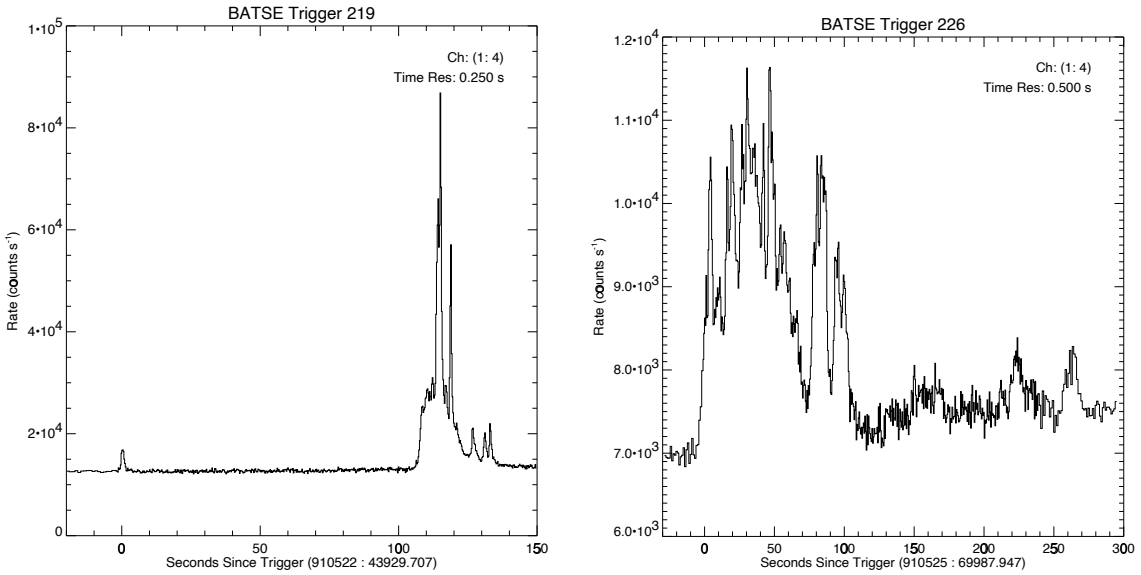



Figure 2.1: Distribution of $T_{90}$-values for 222 GRBs from the first BATSE catalog [33, 54].

Figure 2.1 shows the distribution of $T_{90}$ values for 222 GRBs from the first BATSE catalog. The distribution exhibits pronounced bimodality, with a logarithmic mean $T_{90}$ at one peak of $0.33\ \text{s} \pm 0.21\ \text{s}$ and another at $26.2\ \text{s} \pm 1.7\ \text{s}$ [54]. This leads to the GRB classification scheme: bursts with $T_{90} < 2$ s are Short GRBs (SGRBs) while $T_{90} > 2$ s are Long GRBs (LGRB) [92].




(a) (b)

Figure 2.2: Light curves of 2 different GRBs observed with BATSE [62].

### 2.1.1 Short-duration GRBs

A study by Fong et al. (2013) [34] investigated UV and optical observations of 22 SGRBs using the *Hubble Space Telescope*. They found that SGRBs occur in both elliptical and disk-like galaxies with active star formation. Moreover, SGRBs tend to have median offsets from the centers of their host galaxies that are approximately 3.5 times larger than those observed for LGRBs. Furthermore, SGRBs are located in the faintest UV/optical regions of their host galaxies. Since bright UV/optical regions are associated with active star formation and the presence of massive stars, this suggests that SGRB progenitors migrate from star-forming regions to the eventual explosion sites. Such a spatial distribution provides indirect support for the compact binary merger model due to the natal kicks imparted to neutron stars and black holes [11, 34]. The progenitor of SGRBs is therefore believed to be either a neutron star-neutron star or a neutron star-black hole merger [92]. Such mergers produce gravitational waves. In 2017, the Virgo detectors on LIGO (Laser Interferometer Gravitational-Wave Observatory) observed the gravitational-wave event GW170817, which was emitted by a binary neutron star merger and was associated with the SGRB GRB170817A [3]. This event provided the first direct evidence linking SGRBs to binary neutron star mergers, thereby confirming the compact binary merger hypothesis of the SGRBS progenitors.

### 2.1.2 Long-duration GRB

A common feature of LGRBs is the absence of hydrogen and helium lines in their spectra, suggesting that their progenitors are associated with type Ic supernovae [44, 92]. These supernovae result from the core collapse of massive stars that have lost their outer hydrogen and helium envelopes prior to explosion. This envelope stripping is a characteristic of Wolf-Rayet stars, indicating that LGRBs likely originate from this type of star [90, 91].
The majority of LGRBs are found in irregular, star-forming galaxies, where they tend to occur in the brightest regions[1] [37]. These regions are associated with active star formation,

[1] A few LRGBs have been located in spiral galaxies with active star formation [37].

making the presence of massive stars likely and thereby supporting the massive star origin of LGRBs.

## 2.2 The fireball shock model

In 1986 Goodman and Paczynski [42, 69] proposed the relativistic fireball model to explain the VHE energies and short timescales associated with GRBs. They considered a sudden release of energy into a compact region, leading to the formation of an opaque photon-lepton fireball through extensive electron-positron pair production. This fireball undergoes a relativistic expansion and becomes optically thin once its energy falls below the threshold of pair production. However, the original model could not account for the observed non-thermal spectra and rapid temporal variability of the GRBs. These limitations were addressed by Shemi and Piran [81], who introduced a baryonic load to the photon-lepton fireball. This addition enabled the development of the *fireball shock model*, which incorporates both internal and external shock models [64, 77, 93]. Internal shocks, resulting from collisions within the relativistic outflow, explain the variability and non-thermal nature of the prompt emission, while external shocks, arising from the interaction of the outflow with the circumburst medium, are responsible for the longer-lasting afterglow.

**Internal shock model**

The internal shock model explains how photons gain gamma-ray energies from the kinetic energy of an outflow. To see this, consider an opaque fireball composed of photons, leptons, and baryons [73]. The initially opaque, radiation-dominated fireball expands adiabatically due to extreme internal radiation pressure. As it expands, the temperature and pressure decrease and the plasma becomes transparent to photons at the photospheric radius. The photosphere is the surface at which the outflow becomes optically thin to radiation; below this surface, photons are trapped and undergo repeated scatterings, while above the surface, they may escape freely. During this phase, radiation energy is converted into bulk kinetic energy of the plasma. The efficiency of this conversion depends on the baryonic load: a low baryon content allows the outflow to reach ultra-relativistic speeds because the internal energy is converted into kinetic energy of a small amount of baryonic mass [73]. Although leptons are present in greater numbers, the bulk kinetic energy is mainly determined by the inertia of baryons because baryons are much heavier than leptons. With a few baryons to accelerate, the radiation pressure can drive the entire plasma into an ultra-relativistic outflow, forming the GRB jet [80]. In contrast, a high baryonic load leads to lower Lorentz factors, as more energy is spent accelerating the heavier mass, which could result in a sub-relativistic outflow [73].

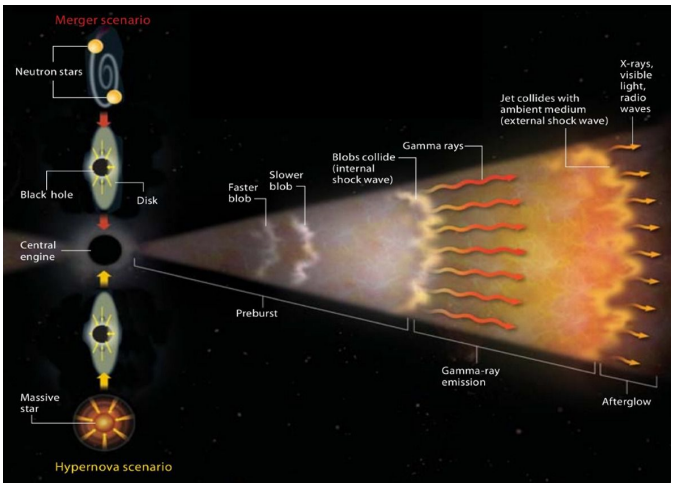



Figure 2.3: Schematic of a GRB from progenitor to afterglow according to the fireball shock model. Figure from [39]

Because the central engine is dynamic, shells of plasma are ejected at varying speeds. When a faster moving shell catches up to a slower one [77], an internal shock is generated; see Figure 2.3. In these shocks, the kinetic energy of the outflow is converted into radiation by accelerated electrons and protons. High-energy electrons can produce gamma-ray photons via *inverse Compton scattering* and *synchrotron radiation* [55, 80]. These mechanisms are described in detail in Section 2.3. The resulting gamma-ray photons produced in the internal shocks constitute the *prompt emission* - i.e., the short-lived, intense flash observed in GRBs [55]. This prompt emission is irregular, and contributes to the highly variable lightcurves of GRBs seen in Figure 2.2.

**External shock model**

When the relativistic shell interacts with the external interstellar medium (ISM) an external shock is formed [73]; see Figure 2.3. It consists of two shocks: a *forward* shock that propagates into the ISM, and a *reverse* shock that travels backward into the ejecta. These shocks are separated by a contact discontinuity [74].

Electrons in both the forward and reverse shocks are accelerated through diffusive shock acceleration (DSA), a process in which particles gain energy by repeatedly crossing the shock front and scattering off magnetic turbulence in the upstream and downstream regions. The DSA results in a power-law distribution of electrons in the downstream region [93]. These accelerated electrons can produce high-energy radiation via both synchrotron emission and inverse Compton scattering, generating an afterglow that spans a broad range of frequencies - from radio to gamma rays [55].

The forward and reverse shocks contribute differently to the observed afterglow. As the forward shock propagates into the ISM, electrons lose energy rapidly through synchrotron emission, leading to a cooling break, which manifests as a steepening in the spectral slope above the cooling frequency [93]. In contrast, the reverse shock travels backward into the ejecta, heating it up and accelerating the electrons. This produces synchrotron radiation that typically peaks in the infrared or optical bands, often resulting in an optical flash [63]. Once the reverse shock has crossed the ejecta, DSA ceases and the electrons cool, leading to a rapid decay of the emission [**\cite {ZHANG_2004}**].

## 2.3 Emission processes in GRBs

The VHE emission observed in GRBs can arise from several radiative mechanisms. Among the most prominent are leptonic processes, in particular *synchtrotron radiation* and *inverse Compton scattering*, which can produce photons with energies extending into the Gev-TeV range. These processes involve relativistic electrons interacting with magnetic fields or upscattering ambient photon fields to higher energies. In addition, hadronic processes, such as photopion production and photopair production, can also contribute to the high-energy photon spectrum. However, these interactions are generally subdominant in GRBs and are primarily relevant in the context of particle cascades. They will not be discussed further in this thesis.

### 2.3.1 Inverse Compton scattering

A process responsible for producing VHE photons in extreme astrophysical events (e.g. GRBs and active galactic nuclei) is inverse Compton scattering [17]. It describes how a low-energy photon gains energy by scattering off a relativistic electron [17], see Figure 2.4.

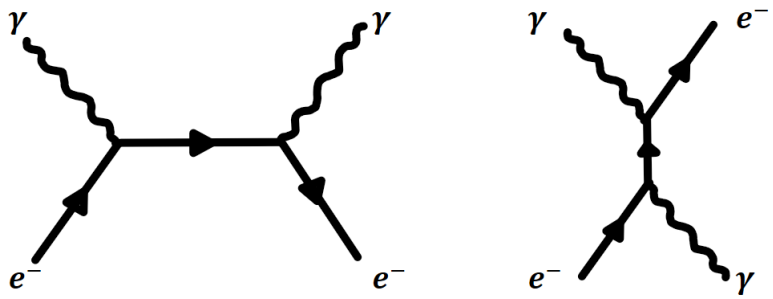



Figure 2.4: The two lowest order Feynman diagrams of Compton scattering.

In the case of GRBs the low-energy photon emerges as black-body radiation from the central engine, while the electrons gain their relativistic speed through DSA [92].

Inverse Compton scattering takes place in two distinct limits called the *Thomson-* and *Klein-Nishina* limits [12]. It is important to distinguish between the two limits as they affect the outgoing photon energies differently. To see this, consider an inverse Compton scattering event in the rest frame of the electron ($K'$), see Figure 2.5. Let $\varepsilon_i'$ and $\varepsilon_o'$ represent the incoming and outgoing photon energies, respectively, in $K'$. $\theta_o'$ is the scattering angle.

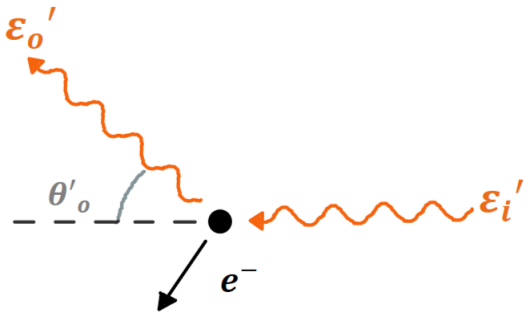



Figure 2.5: Inverse Compton scattering in the rest frame of the electron.

In $K'$, the incoming photons move toward the electron and will appear blue-shifted as a result of a Doppler boost of the photons. As a consequence, the incoming photon energy becomes [78]

$$\varepsilon_i' = \gamma\varepsilon_i \left(1 - \beta\cos\theta_i\right), \tag{2.1}$$

where $\varepsilon_i$ is the incoming photon energy, and $\theta_i$ is the angle between the incoming photon and the electron's direction of propagation, both defined in the lab frame. The Lorentz factor is given by $\gamma = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}$ and $\beta = v/c$, where $v$ is the electron velocity in the lab frame.
Consider Equation (2.1). The energy of the incoming photons as seen from $K'$ is maximized when $\theta_i = \pi$ and minimized for $\theta_i = 0$ with $\varepsilon'_{i,max} = 2\gamma\varepsilon_i$ and $\varepsilon'_{i,min} = \varepsilon/2\gamma$. In a Compton scattering event, the energy of the outgoing/scattered photon in $K'$ is [78]

$$\varepsilon_o' = \frac{\varepsilon_i'}{1 + \left(\frac{\varepsilon_i'}{m_e c^2}\right)\left(1 - \cos\theta_o'\right)} \tag{2.2}$$

The scattered photon energy in the lab frame is obtained by a boost from the rest frame of the electron [78]:

$$\varepsilon_o = \gamma\varepsilon_o' \left(1 + \beta\cos\left(\pi - \theta_o'\right)\right) \approx \gamma\varepsilon_o' \left(1 - \cos\theta_o'\right) \tag{2.3}$$

The energy of the scattered photon in the observer frame is maximized when $\theta_o' = \pi$ with $\varepsilon_{o,max} = 2\gamma\varepsilon'_{o,max}$, which corresponds to a head-on collision. It is evident that $\varepsilon_{o,max}$ depends on the maximal energy of the scattered photon in $K'$. In the Thomson limit Equation (2.2) reduces to $\varepsilon_o' \approx \varepsilon_i'$ while in the Klein-Nishina limit it becomes $\varepsilon_o' \approx m_e c^2/2$. The maximized energies of the scattered photon in the observer frame in each of the limits are as follows:

$$\varepsilon_{o,max} \approx 4\gamma^2\varepsilon_i, \qquad \text{(Thomson)} \tag{2.4}$$

$$\varepsilon_{o,max} \approx \gamma m_e c^2. \qquad \text{(Klein-Nishina)} \tag{2.5}$$

The energy of the scattered photon can increase indefinitely with the energy of the incoming photon in the Thomson limit, but the Klein-Nishina limit does not exhibit this energy dependence. Instead, it is limited by the rest mass of the electron. This difference can be explained by recoil effects on the electron [12]: In the Thomson limit $\varepsilon_i' \ll m_e c^2$, this implies that the recoil of the electron can be neglected, as the scattered photon gains only a small fraction of the electron's energy in each scattering event. In the Klein-Nishina limit, $\epsilon_i' \gg m_e c^2$ leads to a non-negligible recoil of the electron and the electron transfers a large fraction of its energy to the scattered photon. In that sense, the energy loss of the electron is continuous in the Thomson limit, since several scatterings have to occur before it loses a significant fraction of its energy, but it is discrete in the Klein-Nishina limit as it loses a large fraction of its total energy in each scattering event.

In addition to altering the energy of the scattered photon, the Klein-Nishina limit also significantly reduces the scattering cross section, making inverse Compton scattering less efficient at high energies. To see this, consider the Compton cross section [12]:

$$\sigma_{tot} \to \sigma_T \left(1 - \frac{2\varepsilon'}{m_e c^2} + ...\right) .. \tag{2.6}$$

In the Thomson limit, the total cross section reduces to the Thomson cross section, $\sigma_{tot} \to \sigma_T$. However, in the Klein-Nishina limit, quantum corrections become significant, and the cross section decreases with increasing photon energy: $\sigma_{tot} < \sigma_T$, as indicated by the second term in Equation (2.6).
This energy dependence of the cross section directly impacts the observed photon spectrum. Since the spectrum measures the number of scattered photons per unit energy, it naturally depends on whether the scattering occurs in the Thomson or Klein-Nishina limit. To obtain the total inverse Compton spectrum, one integrates over both the electron energy distribution and the incoming photon spectrum [12]. The latter depends on the energy distribution of the initial photons, so integration would be with respect to the incoming photon energies $\varepsilon$ and the gamma factors of the electron distribution $\gamma$ [12],

$$\frac{dN_{tot}}{dtd\varepsilon_1} = \int \int N_e(\gamma)\, d\gamma \left(\frac{dN_{\gamma,\varepsilon}}{dtd\varepsilon_1}\right), \tag{2.7}$$

here $\varepsilon_1$ is the energy of the scattered photons in the lab frame. The photon spectra of the electron and the incoming photon are $N_e(\gamma)$ and $N_{\gamma,\varepsilon}$, respectively. Several spectra of inverse Compton emission can be seen in Figure 2.6 for different $\gamma$.

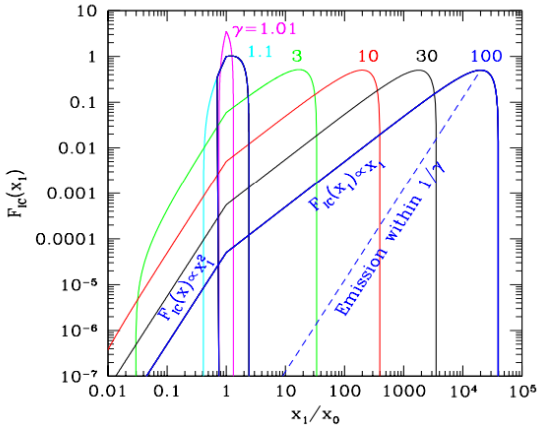



Figure 2.6: Inverse Compton spectrum emitted by scattering of isotropic monochromatic radiation on electrons with different $\gamma$. $x_0$ and $x_1$ denotes dimensionless frequencies of the incoming and scattered photons given as $x = h\nu/m_e c^2$, where $h$ is Planck's constant and $\nu$ is the photon frequency [40].

In the Thomson regime, the inverse Compton spectra follow a power-law if the electron spectrum is a power-law. However, in the Klein-Nishina regime, the suppression of the cross section reduces the number of high-energy scattered photons [40]. This leads to a cut-off of the spectra, as is apparent from Figure 2.6.

### 2.3.2 Synchrotron emission

Synchrotron radiation is produced by relativistic charged particles that are deflected by magnetic fields [17]. These charged particles, typically electrons, follow helical trajectories

around the magnetic field lines. The emission is strongest near the critical frequency $\nu_c$ that is given by [12]

$$\nu_c = \frac{3eB\gamma^2}{4\pi m_e c}, \tag{2.8}$$

where $e$ is the electric charge and $B$ is the magnetic field strength. Synchrotron emission spans a wide range of frequencies, and the frequency of the emitted radiation depends on both the energy of the electrons and the strength of the magnetic field.

Consider a relativistic electron spiralling around the magnetic field lines. The power per unit solid angle emitted by the electron in the direction $\vec{n}$ is [12]

$$\frac{\mathrm{d}P\left(t\right)}{\mathrm{d}\Omega_n} = \frac{e^2}{4\pi c}\frac{\left|\vec{n}\times\left[\left(\vec{n}-\vec{\beta}\right)\times\dot{\vec{\beta}}\right]\right|^2}{\left(1-\vec{n}\cdot\vec{\beta}\right)^6}, \tag{2.9}$$

where $\vec{\beta}$ is the velocity of the electron in the lab frame and $\Omega_n$ is the solid angle. A full calculation of the emitted power involves a Fourier transform on the above equation and integrating over the solid angle. A detailed derivation can be found in [12]; here, only the key results are presented. The spectral power per unit frequency is

$$P\left(\nu\right) = \frac{\sqrt{3}e^3 B}{m_e c^2}\frac{\nu}{\nu_c}\int_{\nu/\nu_c}^{\infty} K_{5/3}\left(x\right)dx \tag{2.10}$$

where $K_{5/3}$ is a modified Bessel function of order 5/3. Figure 2.7 shows the single particle spectrum [17] where

$$F\left(\nu\right) = \frac{\nu}{\nu_c}\int_{\nu/\nu_c}^{\infty} K_{5/3}(x)dx \tag{2.11}$$

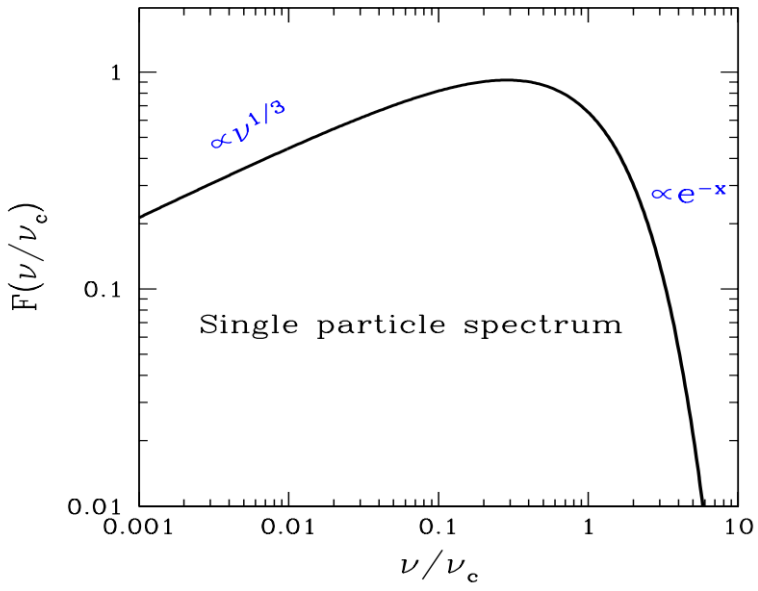



Figure 2.7: Synchrotron spectrum emitted by a single electron [17]

At low frequencies, the synchrotron spectrum of a single electron follows a power-law with spectral index 1/3 [17]:

$$P(\nu) \propto \left(\frac{\nu}{\nu_c}\right)^{1/3} \tag{2.12}$$

At high frequencies, the spectrum exhibits an exponential cutoff [17]:

$$P(\nu) \propto e^{-\nu/\nu_c} \tag{2.13}$$

The total synchrotron spectrum emitted by a distribution of electrons is obtained by integrating the single-electron synchrotron emission over the electron energy distribution [12, 17]. For a power-law distribution of electron energies $N(\gamma) \propto \gamma^{-p}$ [40], the resulting synchrotron spectrum also follows a power-law, with spectral index $(p-1)/2$ [59].
It is important to note that synchrotron emission is beamed in the direction of the electron's motion. Most of the radiation is confined within a narrow cone with an opening angle of $1/\gamma$ [40]. To a distant observer, this beaming effect appears as pulses [17].

## 2.4 Detection of VHE gamma-rays from GRBs

Now that we have introduced how gamma-rays are produced in a GRB, we turn to how they are detected.

### 2.4.1 Air shower

When a *primary* particle - originating from a cosmic ray or VHE gamma ray - enters the atmosphere of Earth, it collides with an atmospheric nucleus (e.g., nitrogen or oxygen), producing several *secondary* particles. Subsequently, these secondary particles interact and decay, initiating a cascade commonly referred to as an *air shower* [43]. The cascade ceases once the particles' energies fall below the threshold for processes such as pair production or pion production, or when they are sufficiently low to be absorbed or scattered without generating additional particles. At this point, the cascade reaches its maximum depth, also known as the *shower* maximum [43].

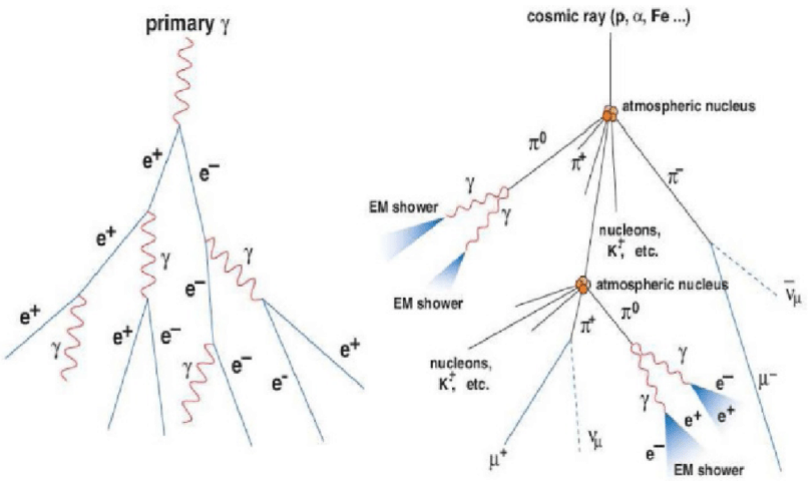



Figure 2.8: Left: VHE photon as primary particle. Right: VHE proton or nucleus as primary particle [60].

Figure 2.8 illustrates two types of air showers: electromagnetic and hadronic [60]. An electromagnetic air shower is typically initiated by a high-energy gamma ray, or occasionally by a VHE electron or positron. When a VHE photon enters the atmosphere, it interacts with the Coulomb field of a nucleus, producing a secondary electron-positron pair through pair production. These secondary particles are then deflected by the electric field of other nuclei, and their acceleration results in the emission of additional gamma rays via bremsstrahlung [5]. This cycle continues until ionization becomes the dominant mechanism of energy loss for the secondary particles [43].
A hadronic shower produces mesons and photons when a high-energy proton or nuclei from a cosmic ray collides with an atmospheric nucleus [43, 49]. Pions are more likely than other mesons to be produced from this collision, although kaons are also found. The neutral pion rapidly decays to $\pi^0 \rightarrow 2\gamma$. The decay product is two gamma-rays and they will undergo pair production upon the interaction with another nucleus, creating a full electromagnetic air shower. The charged pions have a longer decay-time than $\pi^0$ and they produce $\pi^{\pm} \rightarrow \mu^{\pm} + \nu_{\mu}$. The neutrinos and most of the muons from these decays reach the ground- or underground based detectors. For charged kaons the most likely decay modes are $K^{\pm} \rightarrow \mu^{\pm} + \nu_{\mu}$ and $K^{\pm} \rightarrow \pi^{\pm} + \pi^0$. The pions would decay further and the shower continues [5].

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### 2.4.2 LHAASO

LHAASO is an air shower observatory located on Haizi Mountain at an altitude of 4,410m, in Daocheng, Sichuan Province, China [14]. It is designed to detect VHE cosmic rays and gamma-rays using the extensive air showers (EAS) that they produce in the atmosphere. The observatory includes three main instruments: a 1.3 $km^2$ array (KM2A) comprised of electromagnetic particle detectors (EDs) and muon detectors (MDs), a 78,000 $m^2$ water Cherenkov detector array (WCDA) located in the center of KM2A, and 20 wide-field Cherenkov light telescopes (WFCTA) located next to the WCDA [14], see Figure 2.9. Here, I focus on KM2A and WCDA as data from these detectors will be used in this thesis.

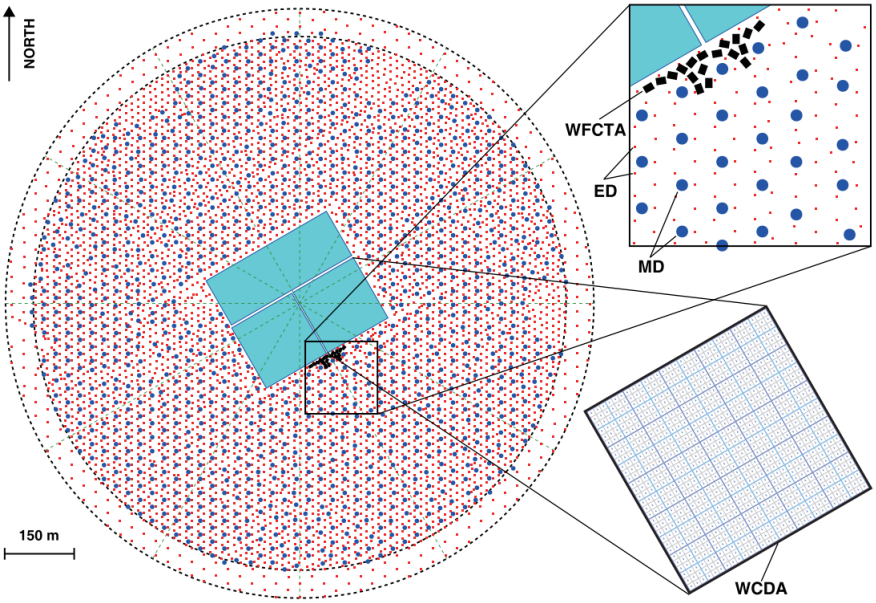
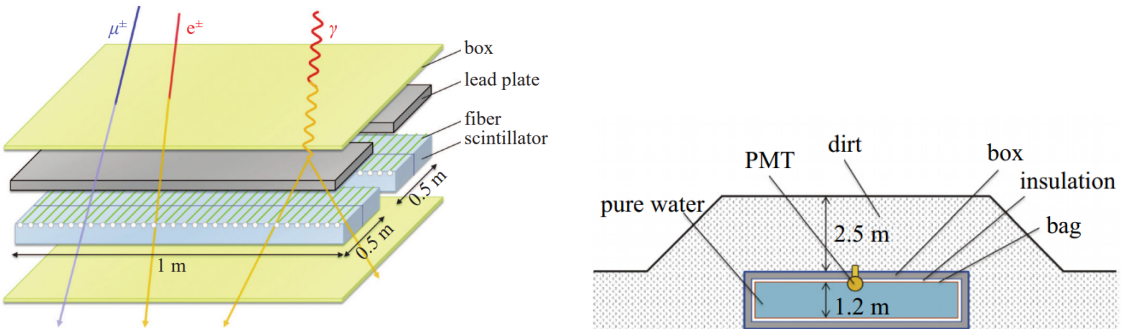



Figure 2.9: The layout of LHAASO [14].

**KM2A**

The KM2A array consists of EDs and MDs, that are uniformly distributed across a circular array. The central array features 4,911 EDs spaced 15m apart, and 1,188 MDs[2] spaced 30m apart. This central array has a radius of 575m. Surrounding it is an outer ring composed of 305 EDs with a 30m spacing. The full array extends to an outer radius of 645m [14]. Each ED has a surface area of $1m^2$ and consists of 4 scintillation tiles that are covered by a 5mm thick lead plate [94], see Figure 2.10a. Low-energy charged particles are absorbed by the lead plate while gamma-rays are converted into electron-positron pairs, improving the angular resolution of the ED array. When the high-energy electron and positron enters the scintillator it excites the scintillating material producing scintillation photons. The scintillation photons are transferred to a photomultiplier where they are transformed into an electric signal [2, 94].




(a) Electromagnetic detector

(b) Muon detector

Figure 2.10: Sketch of electromagnetic detector and muon detector design used at LHAASO [94].

The MDs are pure-water Cherenkov detectors [94]. Each MD consists of a cylindrical

[2]The MDs have a near-uniform distribution. Areas without MDs in the central grid are due to geological conditions that prohibited the construction of MDs [94].

concrete tank with a height of 1.2m and a diameter of 6.8m. A photomultiplier tube is mounted at the top of the tank. The tank is sealed with a lid and buried beneath a 2.5m layer of soil, see Figure 2.10b. This soil layer absorbs most electromagnetic secondary particles[3], while allowing muons to pass through, thereby enabling the MDs to primarily detect muons. When the secondary muons - or, in some cases, very high-energy electrons, positrons, or photons - enter the water, they produce Cherenkov radiation. Cherenkov radiation is detected by the photomultiplier and converted into an electrical signal [2].

**WCDA**

The WCDA consists of three water Cherenkov detector arrays named WCDA-1, WCDA-2, and WCDA-3 [58]. They are essentially squared, pure water pools used to detect secondary particles from the cosmic-ray or gamma-ray induces air shower. Each array is comprised of several unit detectors, which are 5m $\times$ 5m cells at a depth of 4.4m, which are separated by a non-reflective plastic curtain, which prevents light passing through into neighboring cells [94]. Each unit includes a photo multiplier tube pointing upwards. WCDA-1 and WCDA-2 each span an area of 150 m $\times$ 150m and includes 900 detector units, while WCDA-3 is larger, spanning an area of 300m $\times$ 110m [58]. When a secondary particle from an air shower enters the pool, it emits Cherenkov radiation. The photomultiplier measures the Cherenkov radiation, and determines the number of photons and their arrival time [58, 94].

[3]Very high-energy electromagnetic secondary particles could penetrate the soil layer [2].

# Chapter 3

# Axions and axion-like-particles

## 3.1 The strong CP problem

Axion theory emerged as a response to an unresolved problem in quantum chromodynamics (QCD) in the 1970s known as the *strong CP problem* [1]. To understand this, it is useful to first examine the symmetric structure of the QCD Lagrangian for $N_f$ quark flavors [31, 84, 85]:

$$\mathcal{L}_{QCD} = \sum_{f=1}^{N_f} \bar{\psi}_f \left( i\not{D} - m_f \right) \psi_f - \frac{1}{4} F^{a\mu\nu} F^a_{\mu\nu}, \tag{3.1}$$

where $\psi_f$ is the quark field for a quark of flavor $f$, $D_\mu$ is the covariant derivative, and $F^a_{\mu\nu}$ is the gluon field strength tensor:

$$D_\mu = \partial_\mu + igA^a_\mu T^a, \tag{3.2}$$

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu - g f^{abc} A^b_\mu A^c_\nu. \tag{3.3}$$

here $g_s$ is the strong coupling constant, $A^a_\mu$ are the gluon fields, and $T^a$ are the generators of the $SU(3)$ group, with $T^a = \frac{1}{2}\lambda^a$ where $\lambda^a$ are the Gell-man matrices [31].
The QCD Lagrangian has an SU(3) gauge symmetry. This is a color symmetry which means that the QCD Lagrangian is invariant under a local SU(3) transformation of color. When the quark masses vanish (i.e., $m_f \to 0$), the left- and right-handed components of the quark field decouple. This gives rise to a *global* chiral symmetry:

$$U(N_f)_L \times U(N_f)_R. \tag{3.4}$$

Since the up, down, and strange quarks have masses much smaller than the QCD scale ($\Lambda_{QCD} \sim 1$ GeV) [67], it is a reasonable approximation to treat the $N_f = 3$ flavours as effectively massless. This leads to an approximate global $U(3)_L \times U(3)_R$ symmetry. This symmetry group can be decomposed as [84]

$$U(3)_L \times U(3)_R = SU(3)_L \times SU(3)_R \times U(1)_V \times U(1)_A, \tag{3.5}$$

where $SU(3)_L \times SU(3)_R$ corresponds to the global chiral symmetry, $U(1)_V$ is the vector symmetry associated with baryon number conservation, and $U(1)_A$ is an axial symmetry that rotates left- and right-handed quark phases oppositely:

$$\psi_L \to e^{i\alpha}\psi_L, \qquad \psi_R \to e^{-i\alpha}\psi_R. \tag{3.6}$$

The global chiral symmetry group $SU(3)_L \times SU(3)_R$ acts independently on the left- and right-handed quark fields:

$$q_L \mapsto Lq_L \quad \text{and} \quad q_R \mapsto Rq_R, \tag{3.7}$$

where $L$ and $R$ are $3 \times 3$ unitary matrices in $SU(3)_L$ and $SU(3)_R$, respectively. This symmetry is spontaneously broken due to the formation of a non-zero quark condensate in the QCD vacuum,

$$\langle \bar{q}q \rangle = \langle \bar{q}_L q_R + \bar{q}_R q_L \rangle \neq 0, \tag{3.8}$$

which is invariant under vector rotations, i.e., when $L = R$ but not invariant for $L \neq R$. This breaks the chiral symmetry down to the diagonal subgroup $SU(3)_V$, which acts identically on both left- and right-handed fields. Goldstone's theorem implies that for each broken generator of a symmetry, a massless Goldstone boson should appear. Since $SU(3)_L \times SU(3)_R$ has $2 \cdot \left(3^2 - 1\right) = 16$ generators and $SU(3)_V$ has 8, the spontaneous breaking yields 8 Goldstone bosons, which are identified with the light pseudoscalar mesons: $\pi^0$, $\pi^\pm$, $K^0$, $\bar{K}^0$, $K^\pm$, $\eta$ symbreakqcd.
The $U(1)_V$ symmetry corresponds to overall phase rotations of the quark fields

$$q_L \mapsto e^{i\alpha} q_L \quad \text{and} \quad q_R \mapsto e^{i\alpha} q_R, \tag{3.9}$$

and is associated with conservation of the baryon number. It is an exact global symmetry of the QCD Lagrangian and is not broken. Consequently, no Goldstone bosons are expected from the $U(1)_V$ symmetry.
The axial $U(1)_A$ symmetry rotates left- and right-handed quark phases oppositely as $q \mapsto e^{i\alpha\gamma^5} q$:

$$q_L \mapsto e^{i\alpha} q_L \quad \text{and} \quad q_R \mapsto e^{-i\alpha} q_R. \tag{3.10}$$

If the $U(1)_A$ axial symmetry was an exact symmetry of the QCD vacuum, it would imply the existence of parity partners for the hadrons, that is, each hadron would have an axially rotated counterpart with the same mass but opposite parity. This is also known as a parity doublet. However, such doublets are absent in the hadronic spectrum, implying that the $U(1)_A$ symmetry must be broken. If this symmetry were spontaneously broken, it would require a 9th Goldstone boson in addition to the 8 associated with the breaking of $SU(3)_L \times SU(3)_R$. However, only 8 light pseudoscalar mesons are observed. This discrepancy between the number of expected and observed pseudoscalar mesons is known as the $U(1)_A$ *problem* [88].

The resolution of the $U(1)_A$ problem arises from the realization that the QCD vacuum is non-trivial and admits a vacuum angle $\theta$ [1]. This is due to the fact that the QCD vacuum consists of several topologically distinct vacua labeled by an integer winding number $n$. The QCD vacuum is then a superposition of all the vacua [72]:

$$|\theta\rangle = \sum_{n=-\infty}^{\infty} e^{in\theta} |n\rangle . \tag{3.11}$$

If $U(1)_A$ were a symmetry of the QCD Lagrangian, then the axial current $J_5^\mu$ would be conserved, i.e., $\partial_\mu J_5^\mu = 0$, in accordance with Noether's theorem. Although $U(1)_A$ appears

invariant in the classical theory, a non-zero divergence of the axial current is induced by quantum correction [72]s:

$$\partial_\mu J_5^\mu = \frac{g^2 N_f}{16\pi^2} F^a_{\mu\nu} \tilde{F}^{a\mu\nu}. \tag{3.12}$$

This reveals that $U(1)_A$ is not a true symmetry of the QCD Lagrangian but is instead broken by an axial anomaly. Following [72], the anomaly modifies the action resulting in non-zero action within some gauge configurations [1]. This effectively adds a new term to the QCD Lagrangian [72],

$$\mathcal{L}_\theta = \theta \frac{g^2}{32\pi^2} F_a^{\mu\nu} \tilde{F}_{a\mu\nu}. \tag{3.13}$$

This term, or more specifically $F_a^{\mu\nu}\tilde{F}_{a\mu\nu}$, violates parity and time reversal but remains invariant under charge conjugation and must therefore violate CP unless $\theta \to 0$. Even if the QCD Lagrangian has $\theta \to 0$, phases in the complex quark matrix that arise from the weak interactions, would reintroduce CP violation by [28]

$$\bar{\theta} = \theta + \text{Arg det} M, \tag{3.14}$$

where $\bar{\theta}$ would be the parameter entering the $\mathcal{L}_\theta$-term. Its value is constrained by measurements of the electric dipole moment of the neutron (EDMN). The upper limit of the EDMN is currently $10^{-26}$ e cm [8], implying that $\bar{\theta} < 10^{-11}$ [51]. Such a small value of $\bar{\theta}$ is unexplained by the Standard Model and suggests fine-tuning - this is the strong CP problem.

## 3.2 The QCD axion

A solution to the strong CP problem was proposed by Roberto Peccei and Helen Quinn in 1977 [70, 71]. They introduced a global chiral $U(1)$ symmetry, now known as the Peccei-Quinn symmetry $U(1)_{PQ}$, to the QCD Lagrangian. This symmetry is spontaneously broken at an energy scale $f_a$, giving rise to a Nambu-Goldstone boson known as the axion.
The $U(1)_{PQ}$ current is anomalous, giving rise to a non-zero divergence [72]:

$$\partial_\mu J^\mu_{PQ} = \xi \frac{g^2}{32\pi^2} F_a^{\mu\nu} \tilde{F}_{a\mu\nu}, \tag{3.15}$$

where $\xi$ is a model-dependent coefficient. A non-zero divergence of the $U(1)_{PQ}$ chiral current modifies the QCD Lagrangian and as a result, the axion couples to the gluon field strength [72]:

$$\mathcal{L}_{total} = \mathcal{L}_{SM} - \frac{1}{2}\partial_\mu a \partial^\mu a + \mathcal{L}_{int}\left[\frac{\partial^\mu a}{f_a}; \Psi\right] + \left(\bar{\theta} + \xi\frac{a}{f_a}\right)\frac{g^2}{32\pi^2} F_a^{\mu\nu}\tilde{F}_{a\mu\nu}, \tag{3.16}$$

where $\mathcal{L}_{SM}$ refers to the QCD Lagrangian from Equation (3.1), $a$ is the axion field, and $\mathcal{L}_{int}$ describes interactions of axions with other particles of the standard model. The last

term can be interpreted as an effective potential of the axion field, that is minimized when [72]

$$\langle a \rangle = -\frac{f_a}{\xi}\bar{\theta}, \tag{3.17}$$

dynamically cancelling the CP violating terms in the QCD Lagrangian and solving the strong CP problem.
So far, the $U(1)_{PQ}$ symmetry has been treated classically, ignoring non-perturbative QCD effects. Inclusion of these effects generates a non-trivial potential that explicitly breaks the $U(1)_{PQ}$ symmetry, promoting the axion to a pseudo Nambu-Goldstone boson [72]. As a result, the axion acquires a small mass, given by [23]

$$m_a = \frac{m_\pi f_\pi}{f_a}\frac{\sqrt{m_u m_d}}{m_u + m_d} \simeq 5.7\mu\text{eV} \times \left(\frac{10^{12}\text{GeV}}{f_a}\right), \tag{3.18}$$

where $m_\pi$ and $f_\pi$ are the pion mass and decay constant, and $m_u$,$m_d$ are the up and down quark masses respectively. Axions have a two-photon coupling given by [23]

$$g_{a\gamma\gamma} = \frac{\alpha}{2\pi f_a}\left(\frac{E}{N} - 1.92(4)\right) \sim 10^{-13}\text{GeV}^{-1}\left(\frac{10^{10}\text{GeV}}{f_a}\right), \tag{3.19}$$

where $\alpha$ is the fine structure constant and $E/N$ is a model dependent ratio of the electromagnetic and color anomalies. Both the axion mass and couplings scale inversely with $f_a$ implying a strict relation between them. This relation distinguishes the QCD axions from axion-like particles (ALPs), for which mass and couplings to SM particles are independent [18].

The original axion model proposed by Weinberg and Wilczek [87, 89], assumed that the PQ symmetry breaking scale $f_a$ was comparable to the electroweak scale ($\sim$ 250 GeV). This predicted relatively large axion couplings to Standard Model particles, which would have made the axion detectable. However, this axion was not observed, effectively ruling out this model [23].
This led to the development of *invisible axion models* where $f_a \ll 250$ GeV. In these models, the axion is lighter and its couplings to the Standard Model particles are significantly suppressed. The two simplest models are the Kim-Shifman-Vainshtein-Zakharov (KSVZ) [50, 82] and Dine-Fischler-Srednicki-Zhitnitsky (DFSZ) [26, 95] models[1]. In the KSVZ model, the axion is couples to heavy quarks beyond the Standard Model, while the DFSZ model additionally allows for axion couplings to nucleons and leptons. These fermions enter the triangle loop in the left panel of Figure 3.1. Both models predict axion couplings to photons via fermion loops induced by the axial anomaly, while the DFSZ model additionally allows for axion couplings to nucleons [23].

[1]For a thorough review of these models see [23]

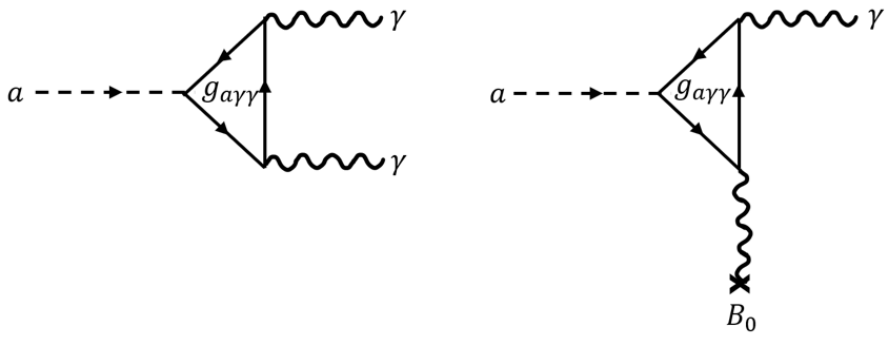



Figure 3.1: Feynman diagrams of the photon-ALP coupling with a coupling strength $g_{a\gamma\gamma}$. The left panel depitcs the decay of an ALP into two photons, while the right panel shows the Primakoff effect, where an ALP converts into a photon in the presence of an external electromagnetic field.

The concept of the axion can be generalized to ALPs that typically also couple to photons but are not constrained by the strong CP problem. In the next section, we explore how ALPs, through their interactions with photons, may provide a framework for explaining the propagation of VHE photons.

## 3.3 Axion-like particles

Although the QCD axion arises specifically as a solution to the strong CP problem, many extensions of the Standard Model predict the existence of ALPs. Like axions, ALPs are light pseudoscalars that couple to photons through a two-photon vertex. However, unlike the QCD axion, ALPs do not exhibit a fixed relationship between their mass and their coupling strength. This decoupling allows ALPs to span a wide parameter space, enabling them to act as cold dark matter candidates [7] or to potentially account for VHE photons observed in certain astrophysical phenomena, such as those associated with GRB221009A [18, 15].
The ALP mass and photon-ALP coupling are collectively referred to as the ALP parameters. Figure 3.2 shows the current limits on these parameters, where the coloured areas correspond to regions of parameter space that have been excluded by experiments or observations. Following the review by [23] the colour coding is defined as follows: The red regions correspond to constraints from terrestrial experiments, such as light-shining-through-wall experiments (e.g., ALPS II [24], OSQAR [9]), helioscopes (e.g., CAST [6]), and laboratory searches. Some of these limits are independent of the assumption that ALPs make up dark matter, while others (e.g., haloscopes like ADMX) assume that the axion or ALP constitutes part or all of the dark matter density.
The green regions represent constraints from astrophysical observations, such as energy loss in stars (e.g. horizontal branch stars, white dwarfs, and supernovae). These limits typically rely on specific model assumptions about ALP production in these environments. The orange band represents the predicted relationship between mass and coupling for QCD axions arising from the KSVZ and DFSZ models. This is not an exclusion region, but rather a theoretical prediction for QCD axions.
The blue region shows cosmological constraints, primarily from considerations of the cosmic axion background and the impact of ALPs on dark matter denisty and cosmology. These limits typically exclude ALPs that are overproduced in the early universe, either as

cold dark matter (via the misalignment mechanism) or as hot relics, and typically constrain the photon-ALP coupling at larger ALP masses ($m_a \gtrsim 10$ eV).

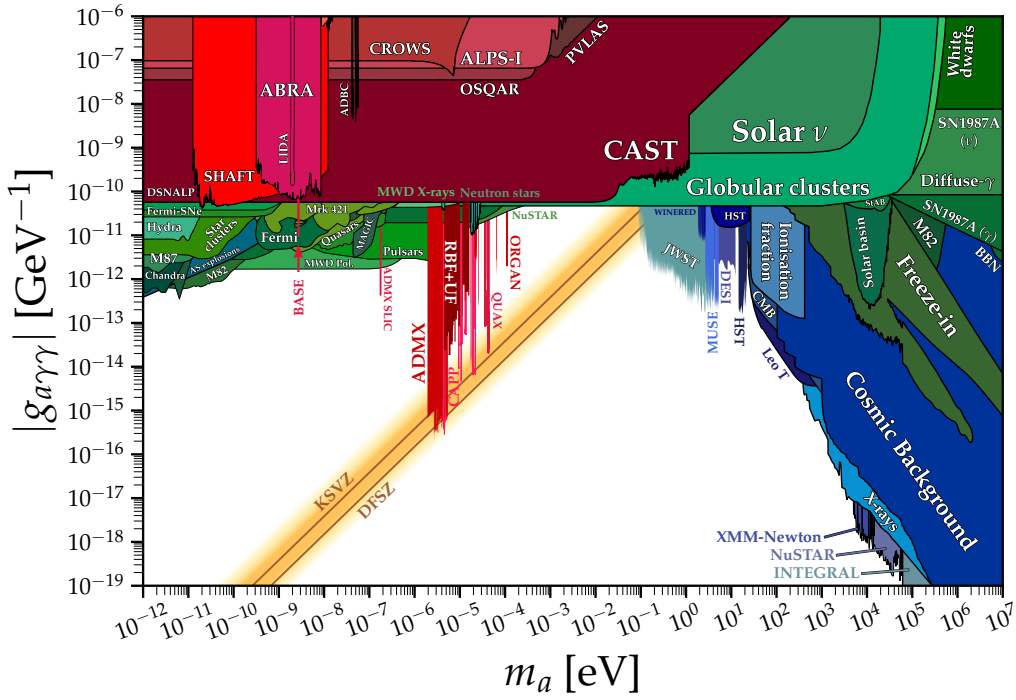



Figure 3.2: ALP-parameter space from [68]. Coloured areas show excluded regions of parameter space from various experimental and observational searches.

For a review on the different experiments used see [47, 23].

### 3.3.1 Photon-ALP interaction

In 1988, Raffelt and Stodolsky [76] showed that photons can mix with low-mass bosons, such as ALPs, in the presence of an external magnetic field via a two-photon vertex. The Lagrangian density describing the photon-ALP system is

$$\mathcal{L} = \mathcal{L}_\gamma + \mathcal{L}_{ALP} + \mathcal{L}_{a\gamma}, \tag{3.20}$$

where $\mathcal{L}_\gamma$ describes photon interactions with the electromagnetic field, $\mathcal{L}_{ALP}$ contains the ALP mass and kinetic terms, and $\mathcal{L}_{a\gamma}$ describes the two-photon vertex which includes photon-ALP interactions. $\mathcal{L}_\gamma$ includes the effective Euler-Heisenberg Lagrangian which describes higher-order corrections to the lowest-order photon-photon interactions, and it arises from vacuum polarizations [76]. These are given by

$$\mathcal{L}_\gamma = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{\alpha^2}{90m_e^4}\left[(F_{\mu\nu}F^{\mu\nu})^2 + \frac{7}{4}\left(F_{\mu\nu}\tilde{F}^{\mu\nu}\right)^2\right], \tag{3.21}$$

$$\mathcal{L}_{ALP} = \frac{1}{2}\left(\partial_\mu a \partial^\mu a - m_a^2 a^2\right), \tag{3.22}$$

$$\mathcal{L}_{a\gamma} = \frac{g_{a\gamma\gamma}}{4}F_{\mu\nu}\tilde{F}^{\mu\nu}. \tag{3.23}$$

Here, $a$ is the ALP field, $m_a$ and $g_{a\gamma\gamma}$ are its mass and coupling to photons, $m_e$ is the electron mass, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field tensor, and $\tilde{F}_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}F^{\rho\sigma}$ is its dual. The entire Lagrangian density uses natural Lorentz-Heaviside units, i.e. $\hbar = c = 1$. In these units, the fine-structure constant is $\alpha = e^2/4\pi$ [76].
To see how photons and ALPs mix under the presence of an external magnetic field, we must start with the equation of motion that describes interactions in the photon-ALP system. It can be derived from the above Lagrangian using the Euler-Lagrange equation:

$$\partial_\mu \left( \frac{\partial \mathcal{L}}{\partial \left(\partial_\mu \phi\right)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0, \tag{3.24}$$

where $\phi = \{a, A\}$ is a single field. Equation (3.20) includes both the ALP field and the photon field, so Equation (3.24) has to be applied to each of the fields. We will neglect the Euler-Heisenberg term for the sake of simplicity. It contributes to refractive indices in the final photon-ALP beam equation. The derivation using the Euler-lagrange equation (without the Euler-Heisenberg term) can be seen in the Appendix A.1.1. Applying the Euler-Lagrangian to Equation (3.20) for the fields $a$ and $A_\nu$ yields the following equation.

$$\left(\partial_\mu \partial^\mu + m_a^2\right) a = \frac{g_{a\gamma\gamma}}{4} \mathbf{E} \cdot \mathbf{B}, \tag{3.25}$$

$$\partial_\mu F^{\mu\nu} = -g_{a\gamma\gamma} \tilde{F}^{\mu\nu} \partial_\mu a + \ldots \quad . \tag{3.26}$$

Expanding $\tilde{F}^{\mu\nu}\partial_\mu a$ from Equation (3.26) into its component form leads to

$$\tilde{F}^{\mu\nu}\partial_\mu a = \left(-\mathbf{B} \cdot \nabla a, \mathbf{B}\partial_t a + \mathbf{E} \times \nabla a\right)^\nu . \tag{3.27}$$

This term can be simplified further by considering the following assumptions: First, the absence of direct ALP sources implies that $\nabla a = 0$. Second, $\tilde{F}^{\mu\nu}$ describes only an external magnetic field. Additionally, by applying Maxwell's equation $\mathbf{E} = -\partial_t \mathbf{A}$ for the photon field and the Lorenz gauge condition $\partial_\mu A^\mu = 0$, Equations (3.25) and (3.26) become

$$\Box \mathbf{A} = g_{a\gamma\gamma} \mathbf{B}_T \partial_t a, \tag{3.28}$$

$$\left(\Box + m_a^2\right) a = g_{a\gamma\gamma} \mathbf{B}_T \partial_t \mathbf{A}, \tag{3.29}$$

where $\mathbf{B}_T$ denotes the transverse external magnetic field.
Consider Equation (3.25). If $\mathbf{E}$ denotes the electric field and $\mathbf{k}$ denotes the wave vector of a photon propagating through space, and if we decompose $\mathbf{B}$ into components perpendicular ($\mathbf{B}_T$) and parallel ($\mathbf{B}_L$) to $\mathbf{k}$, then it is clear from Equation (3.25) that only $\mathbf{B}_T$ survives in the dot product. Therefore, only transversal components of the external magnetic field contribute to photon-ALP mixing.
The vector potential of the photon field, $\mathbf{A}$, is parallel to $\mathbf{E}$ and can be decomposed into two linear photon polarizations: $A_\parallel$ and $A_\perp$ aligned parallel and perpendicular to $\mathbf{B}_T$, respectively. Because $A_\perp \mathbf{B}_T$ only photons linearly polarized along $A_\parallel$ mix with ALPs.

Equations (3.28) and (3.29) can be rewritten by further considering the following: Assume a plane wave with frequency $\omega$ and wavenumber $\mathbf{k}$ propagating in the positive $z$-direction.

For plane waves, we have $\partial_t = i\omega$ which implies that $\Box = \partial_t^2 - \partial_z^2 = -\omega^2 - \partial_z^2$. Applying this definition in Equations (3.28) and (3.29) yields

$$\left(-\omega^2 - \partial_z^2\right) \mathbf{A} = i g_{a\gamma\gamma} \omega \mathbf{B}_t a \tag{3.30}$$

$$\left(-\omega^2 - \partial_z^2 + m_a^2\right) a = i g_{a\gamma\gamma} \omega \left(\mathbf{B}_t \cdot \mathbf{A}\right). \tag{3.31}$$

The decomposition of $\mathbf{A}$ and $\mathbf{B}_t$ into their components and moving all terms to the left-hand side (LHS) of equations (3.30) and (3.31) leads to the following systems of coupled differential equations.

$$\left(-\omega^2 - \partial_z^2\right) A_\perp = 0 \tag{3.32}$$

$$\left(-\omega^2 - \partial_z^2\right) A_\parallel = i g_{a\gamma\gamma} \omega B a \tag{3.33}$$

$$\left(-\omega^2 - \partial_z^2 + m_a^2\right) a = i g_{a\gamma\gamma} \omega B A_\parallel. \tag{3.34}$$

Here, we have defined $|\mathbf{B}_T| = B$.
The idea is to rewrite the above equations of motion into a $3 \times 3$ mixing matrix $\mathcal{M}$. All terms without $\omega^2$ and $\partial_z^2$ in Equations (3.32), (3.33), and (3.34) correspond to the matrix entries $\mathcal{M}_{23}$, $\mathcal{M}_{32}$ and $\mathcal{M}_{33}$,

$$\left[\omega^2 + \partial_z^2 + 2\omega^2 \begin{pmatrix} n_\perp - 1 & n_R & 0 \\ n_R & n_\parallel - 1 & \frac{i g_{a\gamma\gamma} B}{2\omega} \\ 0 & \frac{i g_{a\gamma\gamma} B}{2\omega} & -\frac{m_a^2}{2\omega^2} \end{pmatrix}\right] \begin{pmatrix} A_\perp \\ A_\parallel \\ a \end{pmatrix} = 0. \tag{3.35}$$

The entries with $n_i - 1$ and $n_R$ are inserted manually and account for photons interacting with different media. The terms $n_i - 1$ represent deviations of the refractive index from the vacuum and originate primarily from two contributions: photons propagating in a cold plasma acquire an effective mass and QED vacuum polarization. The latter effect is negligible for weak magnetic fields [18]. The terms $n_R$ arise from the Faraday effect when the photons propagate through optically active media [75].
This equation can be simplified by expanding $\omega^2 + \partial_z^2$ and applying the short-wavelength approximation [18, 75, 76], :

$$\omega^2 + \partial_z^2 = \left(\omega + i\partial_z\right)\left(\omega - i\partial_z\right) \approx \left(\omega + k\right)\left(\omega + i\partial_z\right). \tag{3.36}$$

I am adopting the convention from Peskin & Shroeder [83] where $\vec{k} = -i\vec{\nabla}$. Therefore the factor $(\omega - i\partial_z)$ reduces to $(\omega + k)$. Applying the dispersion relation $k = n\omega$ and assuming that $|n - 1| \ll 1$, this expansion becomes

$$\omega^2 + \partial_z^2 = 2\omega\left(\omega + i\partial_z\right). \tag{3.37}$$

Equation (3.35) can be linearized by applying the short-wavelength approximation above. The terms associated with Faraday rotation can be neglected here, since we are working with photon energies much larger than the ALP mass [18],

$$\left[\omega + i\partial_z + \begin{pmatrix} \Delta_\perp & 0 & 0 \\ 0 & \Delta_\parallel & \Delta_{a\gamma} \\ 0 & \Delta_{a\gamma} & \Delta_a \end{pmatrix}\right] \begin{pmatrix} A_\perp \\ A_\parallel \\ a \end{pmatrix} = 0, \tag{3.38}$$

where the entries are defined as [10, 66]:

$$\Delta_\parallel \equiv \Delta_{pl} + \frac{7}{2}\Delta_{QED} \tag{3.39}$$

$$\Delta_\perp \equiv \Delta_{pl} + 2\Delta_{QED} \tag{3.40}$$

$$\Delta_{a\gamma} \equiv \frac{g_{a\gamma\gamma}B}{2} \tag{3.41}$$

$$\Delta_a \equiv -\frac{m_a^2}{2\omega} \tag{3.42}$$

$$\Delta_{pl} \equiv -\frac{\omega_{pl}^2}{2\omega} \tag{3.43}$$

$$\Delta_{QED} \equiv \frac{\alpha\omega}{45\pi}\left(\frac{B}{B_{cr}}\right)^2. \tag{3.44}$$

Here, $\alpha$ is the fine structure constant, $\omega_{pl}^2 = 4\pi\alpha n_e/m_e$ is the plasma frequency of the medium, $n_e$ is the electron density of the medium, and $B_{cr} \equiv m_e^2/e \simeq 4.41 \cdot 10^{13}$ G [10] is the critical magnetic field.

The equation of motion in Equation (3.38) takes the form of a three-state Shrödinger-like equation, where the spatial coordinate $z$ plays the role of time, and the effective Hamiltonian is given by $H = -(\omega + \mathcal{M})$. To solve the equation and understand the origin of photon-ALP oscillations, we employ the transfer matrix approach described in [10, 18]. Since the transfer matrix associated with $H$ can be factorised as

$$T(z,z_0) = e^{i\omega(z-z_0)}\mathcal{T}(z,z_0), \tag{3.45}$$

where $\mathcal{T}(z,z_0)$ is the transfer matrix of the reduced Shrödinger-like equation,

$$\left(i\frac{\mathrm{d}}{\mathrm{d}z} + \mathcal{M}\right)\psi(z), \tag{3.46}$$

with $\psi(z)$ being the three-state wave function including both photon polarizations and the ALP field, it suffices to diagonalize $\mathcal{M}$. This can be achieved via the similarity transformation

$$\mathcal{D} = \mathcal{U}\mathcal{M}\mathcal{U}^\dagger, \tag{3.47}$$

where

$$\mathcal{U} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}, \tag{3.48}$$

and $\mathcal{D} = \mathrm{diag}(\lambda_1,\lambda_2,\lambda_3)$ with $\lambda_i$ being the eigenvalues of $\mathcal{M}$, and $\theta$ is the photon-ALP mixing angle, given by [10]

$$\theta = \frac{1}{2}\arctan\left(\frac{2\Delta_{a\gamma}}{\Delta_{\parallel} - \Delta_a}\right). \tag{3.49}$$

The eigenvalues of $\mathcal{M}$ are given by [10]

$$\lambda_1 = \Delta_{\parallel} \tag{3.50}$$

$$\lambda_2 = \frac{1}{2}\left(\Delta_{\parallel} + \Delta_a - \Delta_{osc}\right) \tag{3.51}$$

$$\lambda_3 = \frac{1}{2}\left(\Delta_{\parallel} + \Delta_a + \Delta_{osc}\right) \tag{3.52}$$

with

$$\Delta_{osc} \equiv \sqrt{\left(\Delta_{\parallel} - \Delta_a\right)^2 + 4\Delta_{a\gamma}^2}. \tag{3.53}$$

The transfer matrix associated with Equation (3.46) can be written in the basis of the eigenvectors of $\mathcal{M}$ [18]:

$$\mathcal{T}\left(z,z_0;0\right) = e^{i\lambda_1(z-z_0)}\mathcal{T}_1(0) + e^{i\lambda_2(z-z_0)}\mathcal{T}_2(0) + e^{i\lambda_3(z-z_0)}\mathcal{T}_3(0) \tag{3.54}$$

where $\mathcal{T}_i$ are the transfer matrices associated with the eigenstates of $\mathcal{M}$.
To compute the photon-ALP conversion probability, we must evolve the initial photon state $\left|A_{\parallel}\right\rangle$ to a position $z$ by applying the transfer matrix $\mathcal{T}(z,0;0)$, and then project the resulting state onto the ALP state $|a\rangle$. The conversion probability $P_{\gamma\to a}$ is then the square of this transition amplitude:

$$P_{\gamma\to a}(z) = \left|\langle a|\,\mathcal{T}\left(z,0;0\right)\left|A_{\parallel}\right\rangle\right|^2 = \sin^2 2\theta \sin^2\left(\frac{\Delta_{osc}z}{2}\right) \tag{3.55}$$

where $\theta \in [0, \pi/4]$. The mixing is maximal when $\theta = \pi/4$ while no mixing occurs when $\theta = 0$. The angle is given by [18]. Note that $\Delta_{osc}$ appears as an oscillation wavenumber in Equation (3.55) which implies that the oscillation length is

$$L_{osc} = \frac{2\pi}{\Delta_{osc}} \tag{3.56}$$

and its energy dependence will be discussed at the end of this section.
Up to this point, the analysis has assumed a known photon polarization and a constant magnetic field direction along the $z$-axis. However, in realistic astrophysical environments, the magnetic field generally varies in both strength and orientation across different regions, known as domains (e.g., galactic or intergalactic domains). Moreover, the polarization of VHE photons is not directly measurable. Consequently, the photon-ALP beam must be described by a polarization density matrix [10, 18]:

$$\rho = \begin{pmatrix} A_{\perp} \\ A_{\parallel} \\ a \end{pmatrix} \otimes \begin{pmatrix} A_{\perp} & A_{\parallel} & a \end{pmatrix}^{\star} \tag{3.57}$$

The evolution of the system is governed by the Von-Neumann-like equation associated with Equation (3.46),

$$i\frac{\mathrm{d}\rho}{\mathrm{d}z} = [\rho, \mathcal{M}]\,. \tag{3.58}$$

The solution is obtained by propagating the photon-ALP beam from $z_0$ to $z$:

$$\rho\,(z) = \mathcal{T}(z)\rho\,(z_0)\,\mathcal{T}^{\dagger}\,(z)\,, \tag{3.59}$$

To account for the changing orientation of the magnetic field across different domains, we fix the coordinate system such that $\mathbf{B}_T$ is aligned with the $y$-axis. In a generic domain, the angle between $\mathbf{B}_T$ and the $y$-axis is $\varphi$, where $0 \leq \varphi < 2\pi$. This angle can vary from one domain to another as the photon-ALP beam propagates through them. To incorporate this effect, we rotate the mixing matrix in the $x-y$ plane so that $\mathbf{B}_T$ aligns with the $y$-axis in each domain. This is achieved by performing another similarity transformation [10],

$$\mathcal{M}' = V^{\dagger}\,(\varphi)\,\mathcal{M}V\,(\varphi)\,, \tag{3.60}$$

$$V\,(\varphi) = \begin{pmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3.61}$$

This similarity transformation is performed for each variation in $\varphi$. A photon-ALP beam may propagate through several domains, possibly altering $\varphi$ in each domain. For the $k$-th domain the transfer matrix is rewritten as

$$\mathcal{T}_k = V^{\dagger}\,(\varphi_k)\,\mathcal{T}V\,(\varphi_k)\,. \tag{3.62}$$

For propagation through $N$ consecutive domains, the total total transfer matrix becomes an ordered product of the individual transfer matrices in each domain [10]:

$$\mathcal{T}\,(\varphi_N,...,\varphi_1) \equiv \prod_{k=1}^{N} \mathcal{T}(\varphi_k) = \mathcal{T}_N\mathcal{T}_{N-1}...\mathcal{T}_1 = \mathcal{T}_{tot} \tag{3.63}$$

The order of the transfer matrices is crucial: the transfer matrix closest to the density matrix of the initial state must be the transfer matrix of the first domain, since the photon-ALP beam propagates through that domain first. If $\rho_i$ denotes the initial state of a photon-ALP beam, then the probability to find the beam in state $\rho_2$ after a distance $z$ is [18]

$$P_{\rho_1\to\rho_2}\,(z) = \mathrm{Tr}\left(\rho_2\mathcal{T}_{tot}\rho_1\mathcal{T}_{tot}^{\dagger}\right). \tag{3.64}$$

For an initially unpolarised photon beam $\rho_0 = \mathrm{diag}\left(\frac{1}{2},\frac{1}{2},0\right)$ and the probability of observing a photon, of any polarisation, is [66]

$$P_{\gamma\gamma} = \sum_{i=\{1,2\}} \mathrm{Tr}\left(\rho_{ii}\mathcal{T}_{tot}\rho_0\mathcal{T}_{tot}^{\dagger}\right) \tag{3.65}$$

where $\rho_{11} = \mathrm{diag}\,(1,0,0)$, $\rho_{22} = \mathrm{diag}\,(0,1,0)$. Equation (3.65) is the photon survival probability, which is a crucial quantity in high-energy astrophysics. It describes the probability that an initial photon remains detectable as a photon after propagating through a region where photon-ALP conversions may occur.

It is possible that when photon-ALP conversion occurs that the ALP never reconverts to a photon. Because ALP interact weakly with other particles they would go undetected, effectively reducing the observed photon count. Therefore, the photon survival probability captures the net effect of photon-ALP-photon oscillations.
The photon-ALP conversion probability is tied strongly to the *critical energy* [10, 18, 19]. In the following I will use $E$ and $\omega$ interchangeably.
Using Equations (3.49) and (3.55) the photon conversion probability can be expressed as

$$P_{\gamma\to a}(z) = \left(\frac{g_{a\gamma\gamma}B}{\Delta_{osc}}\right)^2 \sin^2\left(\frac{\Delta_{osc}z}{2}\right) \tag{3.66}$$

which reveals that the conversion probability is maximized when $\Delta_{osc} \simeq g_{a\gamma\gamma}B$. This implies that the oscillation length depends inversely on $g_{a\gamma\gamma}$. At the maximal conversion probability, the critical energy is defined as [18]

$$E_{crit} = \frac{\left|m_a^2 - \omega_{pl}^2\right|}{2g_{a\gamma\gamma}B}. \tag{3.67}$$

For photon energies $E \gg E_{crit}$, the system enters the *strong mixing regime*, where the conversion probability becomes energy independent. Below this lower bound, for $E < E_{crit}$, the conversion probability and the oscillation length become energy dependent since $\Delta_{osc} \simeq \left|m_a^2 - \omega_{pl}^2\right|/2E$ and the mixing becomes vanishingly small [19].
At very high energies, the conversion probability is damped as a result of QCD vacuum polarization effects. This sets an upper bound on the strong mixing regime [19]:

$$E_{max} = \frac{90\pi g_{a\gamma\gamma}}{7\alpha}\frac{B_{cr}^2}{B}. \tag{3.68}$$

The strong mixing regime is therefore defined by the energy range $E_{crit} < E < E_{max}$.

# Chapter 4

# Extragalactic background light

The EBL is the collective electromagnetic radiation emitted by stars, gas, and dust since the epoch of recombination. It accounts for most of the diffuse background light from resolved and unresolved sources observed at wavelengths between 0.1 $\mu$m to 1000 $\mu$m excluding the the Cosmic Microwave Background (CMB). The EBL Spectral Energy Distribution (SED) at $z = 0$ is shown in Figure 4.1 different EBL models [79]. The SED exhibits two distinct peaks: a UV/optical peak at $\sim 1$ $\mu$m, primarily from stellar emission of young young and massive stars, and a far-IR peak at $\sim 200$ $\mu$m, arising from dust grains heated by stellar light, which re-emit the radiation at longer wavelengths [29].

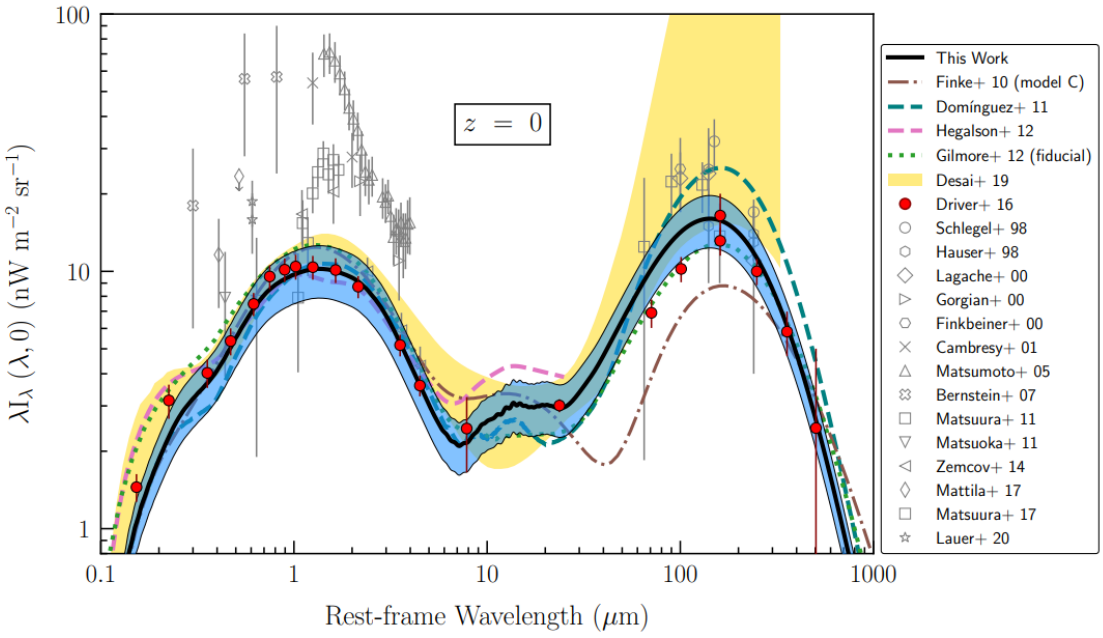



Figure 4.1: Spectral energy distribution of the EBL at $z = 0$, based on various models and observations. The black solid line and blue shaded region represent the EBL estimate from Saldana-Lopez et al. (2021) [79] and its $1\sigma$ standard deviation.

The EBL plays a crucial role in high-energy astrophysics, particularly for understanding the spectra of GRBs and other VHE sources. High-energy gamma-ray photons from distant sources can interact with lower-energy EBL photons via pair production; $\gamma + \gamma_{EBL} \rightarrow e^+ + e^-$, leading to attenuation of the observed flux. The observed photon flux $\phi_{obs}\,(E)$ is then related to the intrinsic flux $\phi_{int}\,(E)$ of the source as [29]:

$$\phi_{obs}\left(E\right) = \phi_{int} e^{-\tau_{\gamma\gamma}\left(E_\gamma, z\right)} \tag{4.1}$$

where $\tau_{\gamma\gamma}\left(E,z\right)$ is the optical depth of the high-energy photons from the source, which depends on both the redshift of the source $z$ and the photon energy $E$. The expression for $\tau_{\gamma\gamma}$ is [29]:

$$\tau_{\gamma\gamma}\left(E_\gamma,z\right) = \int_0^z \mathrm{d}z' \frac{\mathrm{d}\ell}{\mathrm{d}z'} \int_{-1}^{1} \mathrm{d}\mu \frac{1-\mu}{2} \int_{\varepsilon'_{th}}^{\infty} \mathrm{d}\varepsilon n_\varepsilon\left(\varepsilon,z'\right)\left(1+z'\right)^3 \sigma_{\gamma\gamma}\left(\beta',z'\right) \tag{4.2}$$

where $\ell$ is the proper distance, $\mu = \cos\theta$ with $\theta$ being the angle between the two photons, $n_\epsilon$ is the number density and $\epsilon$ is the energy of the EBL photons, $\sigma_{\gamma\gamma}$ is the scattering cross section for two photons and it depends on both the redshift and the parameter $\beta = \left(1 - \epsilon_{th}/\epsilon\right)^{1/2}$. The threshold energy for pair production is given by [29]

$$\epsilon_{th}\left(E_\gamma, \mu, z\right) = \frac{2\left(m_e c^2\right)}{E_\gamma\left(1-\mu\right)} \tag{4.3}$$

where $m_e$ is the electron mass.
The optical depth is sensitive to the assumed EBL model, as different models predict different photon number densities $n_\epsilon$. Since direct measurements of the EBL are hindered by foreground contamination from zodical light (sunlight scattered by interplanetary dust) and galactic emission, various indirect methods have been developed. These include semi-analytic models of galaxy evolution [32, 41], empirical reconstructions from observed luminosity functions [35, 53], and hybrid empirical models constrained by multiwavelength observations [79]. For a review of these and other EBL modeling approaches, see [29]. Each approach makes different assumptions about galaxy formation, stellar populations, and dust content leading to a range of EBL models used in this thesis. The EBL primarily affects photons in the energy range $\sim$ 10 GeV to $\sim$ 10 TeV [29]. At energies above $\sim$ 10 TeV, the universe becomes effectively opaque, and VHE photons from distant sources like GRBs are unlikely to reach Earth [29].

Given the uncertainties in the EBL, I initially considered a range of models: Finke et al. (2022) [32], Franceschini et al. (2017) [35], Dominguez et al. (2011) [27], Gilmore et al. (2012) [41], Kneiske & Dole (2010) [53], and Saldana-Lopez et al. (2021) [79] (including the best-fit, upper- and lower bounds of the SED). These models span a broad range of prediction of the EBL intensity and spectral shape, reflecting the different model assumptions. To determine which models are consistent with the observed photon flux from GRB221009A, I fitted the SEDs of each model to the data and evaluated their p-values. Based on this analysis, half of the models could be rejected, and only the models by Franceschini et al. (2017), Finke et al. (2022), and Saldana-Lopez et al. (best-fit and upper bound) remained compatible with the observations. The following sections provide a brief review of these selected models.

## 4.1 EBL model by Finke et al.

The EBL model developed by Finke et al. (2022) [32] uses a semi-empirical approach to construct the EBL energy density from the ultraviolet (UV) to the far-infrared (far-IR), incorporating both direct stellar emission and the re-emission of absorbed starlight

by interstellar dust. The model is based on the star formation density, dust attenuation, blackbody emission models for both stars and dust, and the initial mass function, which describes the distribution of stellar masses right after their formation.
The stellar component includes both main sequence and post-main sequence stars, all assumed to emit blackbody radiation. A fraction of this light is absorbed by interstellar dust and re-emitted at longer wavelengths in the IR. The model includes three dust components: large grains that absorb optical light and re-emit in the far-IR (temperature $\sim 40$ K), small grains that absorb far-UV light and re-emit in the near-IR ($\sim 70$ K), and polycyclic aromatic hydrocarbons, which produce broad emission features at higher temperatures ($\sim 450$ K). Each dust component is treated as a blackbody emitter, and the total infrared emission is modeled as the sum of these contributions.
The total EBL energy density is obtained by integrating the combined stellar and dust luminosity densities over cosmic time. Although the model extends to high redshifts, its results are considered to be more reliable at low redshift. It predicts a comparatively low gamma-ray opacity at TeV energies.

## 4.2 EBL model by Franceschini et al.

The EBL model by Franceshini et al. (2017) [35] adopts a *backward evolution* model to estimate the background photon density as a function of energy and redshift. This model is based on the number counts, redshift distributions, and luminosity functions of galaxies derived from multi-wavelength observations, combining data from both ground-based observatories (in the optical, near-infrared, and sub-milimeter bands) and space-based telescopes such as Spitzer and Herschel. Based on these observations, the luminosity functions of galaxy populations are extrapolated backward in cosmic time to reconstruct the background photon density.
The 2017 update to the original model [36] incorporated new data from the Herschel Space Observatory and the Spitzer Space Telescope, allowing for an improved determination of number counts and luminosity functions, particularly in the infrared regime. Since the optical and near-infrared parts of the galaxy spectra remained largely unchanged, the main revisions affected the mid- and far-infrared wavelength range ($\lambda > 8$ $\mu$m). This led to slightly lower estimates of the EBL intensity in these regions compared to the original model. However, the updated model still predicts a higher opacity for TeV photons than the model by Finke et al. (2022).

## 4.3 EBL model by Saldana-Lopez et al.

The EBL model by Saldana-Lopez et al. (2021) [79] is constructed using multiwavelength observations from the UV to far-IR of approximately 150,000 galaxies, detected up to $z \sim 6$. The goal of this model is to reduce the uncertainties associated with the evolution of the EBL SED and to improve the modeling of the far-IR region of the EBL. The evolution of the EBL SED is reconstructed by stacking the SEDs of individual galaxies and calculating the total luminosity density in different redshift bins. The far-IR region is addressed by assigning a dust emission component in the mid- and far-IR to every galaxy in the sample. This accounts for the lack of far-IR detection in $\sim 93.8\%$ of the galaxies.
Since the model relies entirely on observational data, the resulting EBL SED includes a best-fit model with associated $1\sigma$ uncertainties. This provides three EBL models: the best fit, a lower bound, and an upper bound. Each model predicts different levels of attenuation for VHE photons: the lower bound corresponds to a lower EBL density and is therefore

less opaque to VHE photons, while the upper bound corresponds to a higher EBL density and stronger attenuation of VHE photons.

# Chapter 5

# GRB221009A

GRB221009A is the brightest GRB ever observed. It was detected on October 9th, 2022, at 13:16:59 Universal Time (denoted as the trigger time $t_0$) by the Fermi Gamma-ray Burst Monitor (GBM) [86]. Approximately one hour later, the Burst Alert Telescope (BAT) onboard the *Swift* satellite also detected the burst, followed by observations with the X-ray Telescope (XRT) and the Ultraviolet and Optical Telescope (UVOT). The GRB was localized at $RA_{J2000} = 288.26452$ and $DEC_{J2000} = 19.77350$, with a 90%-confidence error radius of 0.61 arcseconds [25].
Follow-up spectroscopy with X-shooter at the Very Large Telescope (VLT), 11.55 hours after the trigger, determined the redshift to be $z = 0.151$ [20], corresponding to a luminosity distance of approximately 753 Mpc [45]. The total emitted isotropic-equivalent energy is estimated at $E_{iso} \sim 2 \cdot 10^{54}$ erg, placing GRB221009A among the most energetic GRBs ever recorded.
In addition, GRB221009A was detected by LHAASO using its KM2A and WCDA detectors [13, 15], capturing both the prompt emission and subsequent aftlerglow [13]. More than 5000 photons with energies above 0.5 TeV were reported within 2000s after the trigger [45], with reconstructed photon energies reaching up to 13 TeV. The WCDA, which is sensitive in the $0.2 - 7$ TeV range, reported a detection significance above $100\sigma$, while KM2A, covering $3 - 10$ TeV, reported a significance of $10\sigma$. These detections mark the highest photon energies ever observed from a GRB.
The light curves from KM2A and WCDA span the time interval from $t_0 + 230$s to $t_0 + 900$s [13, 15] and this interval is divided into two segments: $t_0 + 230$s to $t_0 + 300$s, during which the emission begins, peaks and falls off quickly, and $t_0 + 300$s to $t_0 + 900$s, where the emission gradually declines and fades [15]. LHAASO reconstructed the SED and photon energies separately for these two time intervals, using two different models for the intrinsic spectrum of the GRB. Specifically, they applied a log-parabola (LP) model and a power-law with exponential cutoff (PLEC). The LP model resulted in a reconstruced SED extending up to $\sim$ 18 TeV, whereas the PLEC model yielded a lower maximum photon energy of $\sim$ 13 TeV [15]. Their resulting SEDs using both models are shown in Figure 5.1.

In this thesis, we use data from the KM2A and WCDA detectors (hereafter referred to as LHAASO data), that was reconstructed using the PLEC model $300 - 900$s after the trigger. This dataset is available in [57].
The detection of photons above $\sim$ 10 TeV is unexpected, as such photons are expected to suffer from absorption on the EBL (see Section 4). This discrepancy provides a unique opportunity to probe VHE gamma-ray propagation over cosmological distances and test scenarios such as photon-ALP oscillations.

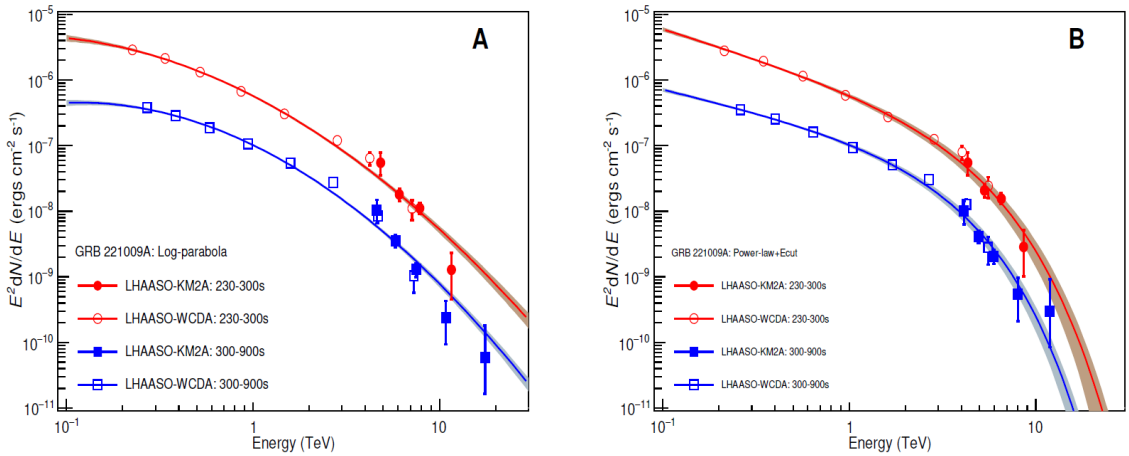



Figure 5.1: Reconstructed SEDs of GRB221009A by LHAASO for $t_0$+230s to $t_0$+300s (red points) and $t_0 + 300$s to $t_0 + 900$s (blue points). The empty squares are WCDA datapoints and the filled squares are KM2A datapoints. The solid line is their best-fit results and the shaded region indicate the $1\sigma$ error region. Plot A uses a log-parabola model while plot B adopts a power-law with eksponential cutoff [15].

A study by Galanti et al. [38] explored whether photon-ALP oscillations could explain the VHE photons observed by LHAASO. They modeled the intrinsic spectrum of GRB221009A as a power-law, based on the spectral shape reported by LHAASO up to 7 TeV, and extrapolated it to 20 TeV. They computed the expected observed spectra under two models: pure EBL attenuation and photon-ALP oscillations. Their results suggest that photon-ALP conversions could alleviate the discrepancy between the VHE photon detection and standard EBL absorption models. However, their analysis is not based on the actual spectrum obtained by LHAASO. It is therefore still necessary to investigate whether the observed spectrum can be explained by a model with pure EBL attenuation or if photon-ALP mixing must be invoked.

# Chapter 6

# Modelling of Observed Spectra

## 6.1 Model of Observed Spectra

The possibility of photon-ALP oscillations in the presence of magnetic fields can modify the observed spectra of GRBs, potentially allowing VHE energy photons to reach Earth. This effect is accounted for by multiplying the intrinsic source spectra $\phi_{int}(E)$ with the photon survival probability:

$$\phi_{obs}(E) = \phi_{int}(E) P_{\gamma\to\gamma} \tag{6.1}$$

where $\phi_{obs}$ is a differential photon flux in units of photons/cm$^2$/s/GeV. The corresponding spectral energy distribution (SED) is then given by multiplying with a factor of $E^2$:

$$\Phi_{model}(E) = \phi_{obs}(E) E^2. \tag{6.2}$$

For a model that includes photon-ALP oscillations and attenuation by the EBL, the photon survival probability is given by Equation (3.65). In contrast, for a model describing only EBL attenuation, the survival probability simplifies to:

$$P_{\gamma\to\gamma} = e^{-\tau_{\gamma\gamma}} \tag{6.3}$$

where the optical depth $\tau_{\gamma\gamma}$ depends on the chosen EBL model, as discussed in Section 4. The intrinsic spectrum is modeled using either a *power-law* (PL) or *log-parabola* (LP):

$$\phi_{int,PL}(E) = \phi_0 \left(\frac{E}{E_0}\right)^{-\alpha} \tag{6.4}$$

$$\phi_{int,LP}(E) = \phi_0 \left(\frac{E}{E_0}\right)^{-\alpha-\beta\ln\left(\frac{E}{E_0}\right)} \tag{6.5}$$

where $E_0$ is a reference energy (set to 1 TeV in this analysis), $\phi_0$ is a prefactor, $\alpha$ is the spectral index, and $\beta$ is a curvature parameter. These models are based on observed spectral shapes rather than derived from fundamental physics, and their origin can be described in the following way:

- The power-law model is motivated by non-thermal emission processes in GRBs, such as synchrotron radiation, which produces a power-law spectrum (if the electrons follow a power-law distribution).

- The log-parabola model is introduced to account for the spectral curvature of GRB spectra in log-space.

One can choose either model for the intrinsic spectrum of a GRB and determine which provides a better fit to the data.

## 6.2 Implementation Details of the Model

### 6.2.1 Parameter Optimization

To obtain a best-fit model of the observed spectrum, we employ a *maximum-likelihood* approach. In this approach, the model parameters $\phi_0$, $\alpha$, and $\beta$ are estimated by maximizing the log-likelihood function. The model that includes these parameters is then most likely to explain the LHAASO data.
For data points with Gaussian-distributed errors, the log-likelihood function is related to $\chi^2$-value [67]:

$$\chi^2(\boldsymbol{\theta}) = -2\ln L(\boldsymbol{\theta}) + \text{const} \tag{6.6}$$

where $\boldsymbol{\theta}$ is a vector containing the model parameters. Thus, maximizing the likelihood is equivalent to minimizing $\chi^2$.
In this thesis, we consider the two following hypotheses. The *null hypothesis* $H_0$: The observed spectrum of GRB221009A, measured by LHAASO, is described by a model based on EBL attenuation. The *alternate hypothesis* $H_1$: The observed spectrum is described by a model that includes photon-ALP mixing and EBL attenuation. To determine which model is preferred by the data, we perform a hypothesis test:

$$\Delta\chi^2 = \chi^2_{EBL} - \chi^2_{ALP} \tag{6.7}$$

where $\chi^2_{EBL}$ and $\chi^2_{ALP}$ are the $\chi^2$ values associated with the models based on EBL attenuation and photon-ALP mixing, respectively.
However, since the errors of the LHAASO data are asymmetric, we have to adopt an asymmetric $\chi^2$, which is defined as

$$\chi^2(E) = \begin{cases} \left(\frac{\Phi_{LHAASO}(E)-\Phi_{model}(E)}{\sigma_p(E)}\right)^2 & , \Phi_{model}(E) > \Phi_{LHAASO}(E) \\ \left(\frac{\Phi_{LHAASO}(E)-\Phi_{model}(E)}{\sigma_n(E)}\right)^2 & , \Phi_{model}(E) < \Phi_{LHAASO}(E) \end{cases} \tag{6.8}$$

where $\Phi_{LHAASO}$ is the SED measured by LHAASO, and $\sigma_{p/n}$ its upper and lower error bars at energy $E$.
When fitting models of the observed spectrum to the datapoints from LHAASO the parameters of the model, that is $\phi_0$, $\alpha$, $\beta$ have to be optimized to obtain the best fit model. Such parameter optimization can be performed using the *Iminuit* [21].

### 6.2.2 Computation of The Photon Survival Probability

The photon survival probability entering Equation (3.65) is computed by evolving the density matrix using transfer matrices for each successive environment, as was outlined in Section 3.3.1. In this work, the photon survival probability of a gamma-ray propagating space from GRB221009A to Earth is calculated using the software `GammaALPs` [65]. To perform

these calculations, GammaALPs requires the redshift and coordinates of GRB221009A, the ALP parameters, the angle between the magnetic field and photon beam, and the magnetic field models of the environments that the photon-ALP beam traverses.
In this work, we consider the following ranges for the ALP parameters:

$$m_a = \mathcal{O}\left(10^{-3} - 10^{3}\right) \text{ neV}, \quad g_{a\gamma\gamma} = \mathcal{O}\left(10^{-12} - 10^{-10}\right) \text{ GeV}^{-1} \tag{6.9}$$

and we set $\varphi = \pi/2$ (the angle between $\mathbf{B}_T$ and the $y-$axis from Section 3.3.1) to maximize the mixing probability. The ALP mass is chosen to be ultralight as efficient photon-ALP mixing requires $m_a \lesssim 10^{-6}$ eV. The upper limit of the photon-ALP coupling is motivated by the upper limit obtained by the CAST helioscope at $|g_{a\gamma\gamma}| < 8.8 \times 10^{-11}$ GeV$^{-1}$ (at 95% CL for $m_a \lesssim 0.02$ eV) [6]. In addition, several astrophysical measurements provide constraints on the ALP parameter space (see Figure 3.2). However, these results depend on modeling assumptions, particularly with regard to magnetic field structures, stellar evolution, and intergalactic propagation, and are therefore subject to uncertainties [67]. Given these considerations, the chosen range of the ALP parameters is still viable for modeling the spectrum of GRB221009A.

The photon-ALP beam propagates through four distinct environments: the GRB source, the host galaxy, extragalactic space, and the Milky Way. The magnetic fields in each environment are modeled following the approach of Galanti et al. [38]. The following review follows Galanti et al. and references therein.

**GRB source**
In this model, the source refers to the shocked fluid in the downstream region of the forward shock produced by the interaction between the GRB ejecta and the ISM (see Section 2.2). Galanti et al. estimate the magnetic field strength in the comoving frame of the shocked fluid and it depends on the particle number density.
They modeled the particle number density in the external medium by combining two components: the ISM contributes with a constant number density, and the stellar wind of the progenitor star contributes with a wind-like profile falling of as $R^{-2}$, where $R$ is the distance from the central engine.
The parameters they adopt are chosen to maximize the photon survival probability, as they aim to assess whether photon-ALP conversion in the source impacts the observed gamma-ray flux under optimistic conditions.

**Host galaxy**
Observations indicate that the host galaxy of GRB221009A is disc-like an oriented edge-on. Since GRB221009A is located close to the nuclear region, Galanti et al. assume that its position is near the galactic center.
They considered two types of host galaxies: a spiral galaxy, modeled with an electron density $n_{host,e} = 1$ cm$^{-3}$, and a starburst galaxy with a higher electron density $n_{host,e} = 1000$ cm$^{-3}$, which reflects its higher star formation activity.
The magnetic field of the host galaxy, $\mathbf{B}_{host}$, is composed of three components:

- A large-scale regular field ($\mathbf{B}_{reg}$)
- A large-scale anisotropic turbulent field ($\mathbf{B}_{aniso}$)
- A small-scale isotropic turbulent field ($\mathbf{B}_{iso}$)

Each component is modeled using an exponential profile in the radial direction of the host galaxy's disk.

For **spiral galaxies**, they include only the regular component $\mathbf{B}_{reg}$ in the model. This is because the lower star formation rate in spiral galaxies leads to a lower supernova explosion rate, making the turbulent components less significant for photon-ALP conversion. Although turbulent fields exist in spiral galaxies, they are subdominant in this context and neglected in the model.
For **starburst galaxies**, all magnetic field components could, in principle, contribute to $\mathbf{B}_{host}$. However, since the average turbulent coherence length is much larger than the coherence length of $\mathbf{B}_{iso}$, the contribution from $\mathbf{B}_{iso}$ is considered negligible compared to that of $\mathbf{B}_{aniso}$ for photon-ALP conversion. Therefore, the model includes only the sum of $\mathbf{B}_{reg}$ and $\mathbf{B}_{aniso}$.

**Extragalactic space**
The extragalactic magnetic field (EGMF) is modeled as a domain-like network. Each domain has a size $L_{dom}^{ext}$ equal to the coherence length of the magnetic field. Within each domain the magnetic field $B_{ext}$ is assumed to be homogeneous, and its direction changes randomly from one domain to the next. Galanti et al. adopt the Domain-Like Smooth-Edges (DLSME) model, to continuously change the magnetic field across the interface between different domains.
Photons propagating through extragalactic space are also attenuated by the EBL, as discussed in Section 4 . However, since ALP do not interact with the EBL a photon that has been converted into an ALP will propagate unimpedded, effectively reducing the optical depth for the photon-ALP beam.

**Milky Way**
Galanti et al. estimate the Galactic Magnetic Field (GMF) of the Milky Way using the model by Jansson and Farrar [48]. This model divides the GMF into three large-scale structural components: the disk, the toroidal halo, and an out-of-plane poloidal component, known as the X-field component, near the galactic center. The magnetic field in each of these regions is composed of a large-scale regular field and a striated random field.
Each of the components is modeled to be divergenceless, and flux conservation is enforced separately for each component. The model parameters are determined by fitting to Faraday rotation measures and polarized synchrotron emission data.
It should be noted that only the regular component is relevant for photon-ALP oscillations with the oscillation lengths considered in this work. The turbulent component would lead to a negligible net contribution of the photon survival probability, due to the rapid conversions that arise in a turbulent magnetic field.

The specific parameters used in this work can be seen in Table 6.1. Although `gammaALPs` supports any of its implemented EBL models, we adopt the four presented in Section 4.

| | GRB source | Host galaxy | | EGMF | GMF |
|---|---|---|---|---|---|
| | | Spiral | Starburst | | |
| $B_0\,[\mu\mathrm{G}]$ | $2 \cdot 10^6$ | 5 | 20 | $10^{-3}$ | – |
| $L_0\,[\mathrm{kpc}]$ | $R/45$ | 10 | 10 | $10^3$ | – |
| $n_0\,[\mathrm{cm}^{-3}]$ | 450 | 1 | 1000 | $10^{-7}$ | $1.1 \cdot 10^{-2}$ |
| $r_{abell}\,[\mathrm{kpc}]$ | $L_0 \cdot 1.1$ | 10.0001 | 10.0001 | – | – |
| EBL model | – | – | – | any | – |
| model | – | – | – | – | jansson12 |

Table 6.1: Parameters used to compute the photon survival probability for a photon-ALP beam propagating through four different environments: the GRB source, the host galaxy, extragalactic space, and the Milky Way. The table also lists the parameters for the two types of host galaxies considered in this work: spiral and starburst. The parameters are from [38]

### 6.2.3 Energy binning around LHAASO datapoints

When comparing the LHAASO datapoints with models of the observed spectra, it is important to account for two key effects: the rapid photon-ALP oscillations and the steep decline in flux from EBL attenuation at high energies.
In models that include photon-ALP oscillations, the predicted photon flux exhibits rapid, small-scale oscillations. Compared directly to the 13 discrete LHAASO datapoints, these oscillatory features may accidentally align with the sparse datapoints, leading to an artificially low $\chi^2$ value and a seemingly good fit. However, this would not reflect a true agreement between the model and the overall trend of the data, but rather an artifact of comparing individual datapoints to a model evaluated at a single energy.
To address this issue for both the photon-ALP mixing and the EBL attenuation models, the data were binned in energy: For each LHAASO datapoint, an energy bin is constructed, and the modeled flux is integrated across that bin. It was necessary to construct the energy bins because LHAASO did not provide the upper and lower limits of their binning. The comparison is then made between the LHAASO datapoint and the average flux of the model within that bin. This approach ensures that the comparison reflects the overall trend of the model rather than coincidental alignments. It also captures the average behavior of the model flux across the energy range represented by each datapoint, making the fit more representative.

**Construction of the energy bins**
Let $E_{cen,i}$ denote the energy of the $i$-th LHAASO datapoint. This is treated as the central energy of the corresponding energy bin (see Figure 6.1). The upper and lower limits of the energy bin, $E_{max,i}$ and $E_{min,i}$, are defined as follows: The upper limit $E_{max,i}$ is the midpoint in log-space between $E_{cen,i}$ and $E_{cen,i+1}$:

$$E_{max,i} = \sqrt{E_{cen,i} \cdot E_{cen,i+1}} \tag{6.10}$$

The bins are adjacent in log-space, such that $E_{max,i} = E_{min,i+1}$. For the first and last datapoints, where the limits are not defined by the neighboring datapoints, the boundaries

are based on the following assumption: $E_{cen,i} = \sqrt{E_{min,i} \cdot E_{max,i}}$, from which we can obtain the lower and upper limits to the first (0) and last ($n$) datapoint, respectively.

$$E_{min,0} = \frac{E_{cen,0}^2}{E_{max,0}}, \qquad E_{max,n} = \frac{E_{cen,n}^2}{E_{max,n}} \tag{6.11}$$

A schematic of this binning method is shown in Figure 6.1.

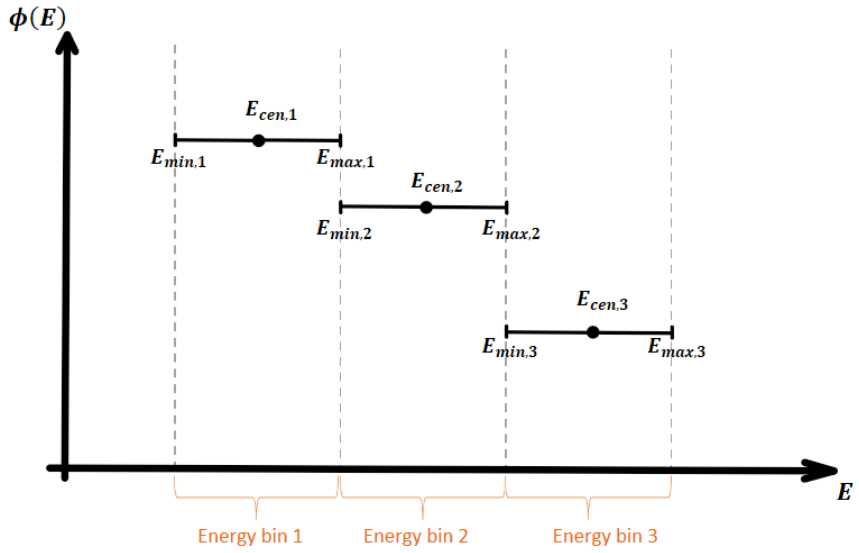



Figure 6.1: Schematic illustration of the energy binning procedure. Each LHAASO datapoint at energy $E_{cen,i}$ is associated with an energy bin defined by the upper and lower limits $E_{max,i}$ and $E_{min,i}$, respectively.

**Averaging procedure and model comparison**

As discussed, it is necessary to average the model flux across each energy bin to ensure a representative comparison between the LHAASO datapoints and the model. The model flux is computed as:

$$\Phi_{obs}\left(E\right) = \phi_{int}\left(E\right) P_{\gamma\to\gamma}\, E^2 \tag{6.12}$$

where $\phi_{int}\left(E\right)$ is the intrinsic flux, $P_{\gamma\to\gamma}\left(E\right)$ is the photon survival probability, and the factor $E^2$ arises from the conversion to the SED. This model is evaluated at a grid of energies with a high resolution.

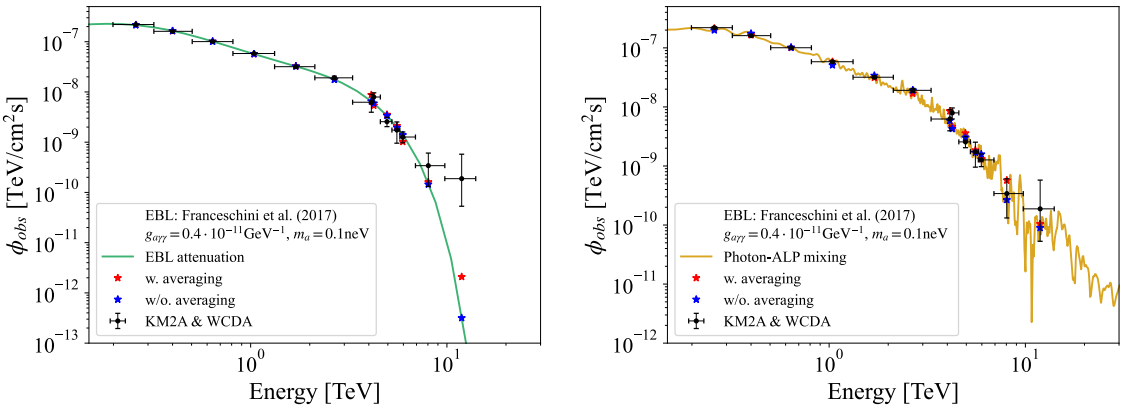
For each energy bin, the average model flux is computed as

$$\Phi_{avg,i} = \frac{1}{\Delta E} \int_{E_{min,i}}^{E_{max,i}} \Phi_{obs}\left(E\right) \mathrm{d}E. \tag{6.13}$$

where $\Delta E = E_{max,i} - E_{min,i}$ which is assumed to be constant. This averaging ensures that the comparison captures the general trend across the bin, rather than being sensitive to individual model values at specific energies. The integration was performed numerically using Simpson's method as implemented in `scipy.integrate.simpson()` [22].
The effect of averaging is illustrated in Figure 6.2, which shows the modeled fluxes for two cases:

- In Figure 6.2a, The model includes only EBL attenuation using the Franceschini et al. (2017) model [35].
- In Figure 6.2b, the model includes both photon-ALP mixing, with $m_a = 0.1$ neV and $g_{a\gamma\gamma} = 0.4 \cdot 10^{-11}$ GeV$^{-1}$, and EBL attenuation.

Both figures display the LHAASO datapoints with vertical error bars, the model flux with averaging (red stars) and without averaging (blue stars). The horizontal error bars of the datapoints represent their energy bins.




(a) Attenuation by the EBL

(b) Inclusion of photon-ALP mixing

Figure 6.2: Comparison of model predictions and LHAASO datapoints. The red stars show the model flux averaged over the energy bins, while the blue stars show the model flux without averaging. The LHAASO datapoints are shown with their vertical error bars, and horizontal error bars indicating their energy bins. Panel (a) shows the model including only EBL attenuation, while panel (b) includes both photon-ALP mixing an EBL attenuation.

For the model including only EBL attenuation (Figure 6.2a), the effect of averaging is primarily visible at energies above $\sim 5$ TeV, where the flux declines rapidly. In particular, the last datapoint shows a significant difference, where the averaged model flux is higher than the regular model flux, resulting in a lower $\chi^2$ contribution from that datapoint.
For the photon-ALP mixing model (Figure 6.2b), the effect of averaging is pronounced throughout the entire energy range. The model's oscillatory behaviour leads to flux points that can lie either above or below the averaged value. The averaging process captures the overall trend of the model, incorporating the oscillations within each bin. Without averaging, the model flux at individual points may not reflect the general behaviour potentially leading to artificially low or high $\chi^2$ values depending on the coincidental alignments.
In summary, the averaging procedure ensures that the comparison between model and data is not biased by the specific behaviour of the model at specific energies. It provides a more accurate assessment of how well the model reproduces the general trend of the observed spectra.

## 6.3 Results: Comparison to LHAASO data

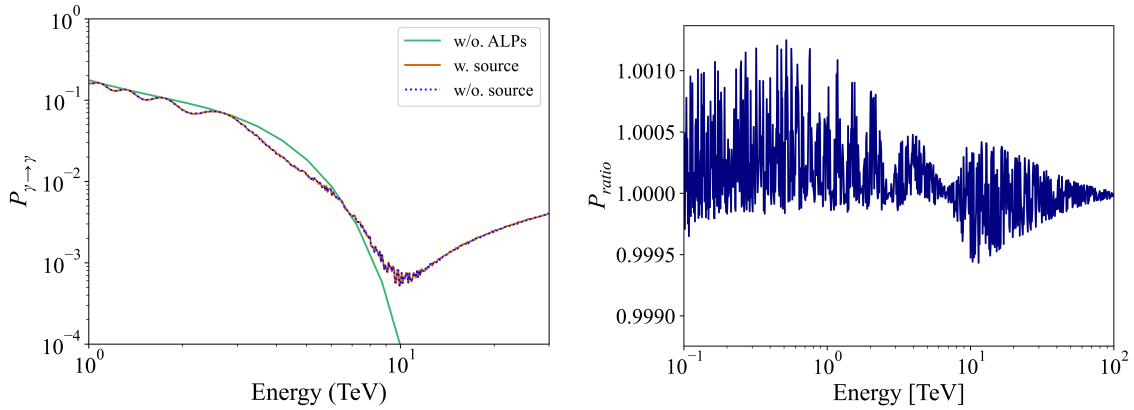
### 6.3.1 Preliminary results

As mentioned earlier our model of the SED depends on both the intrinsic spectrum and the photon survival probabilities. The photon survival probability was dependent on the

environments that the photon-ALP beam propagates through. In Figure 6.3a the photon survival probability is plotted for three different scenarios: attenuation by the EBL (green line), photon-ALP mixing in the source, host galaxy, and the Milky Way (orange line), and photon-ALP mixing in the host galaxy and the Milky Way (blue line), both also includes attenuation by the EBL in intergalactic space. It is clear that photon-ALP mixing enhances the photon survival probabilities at energies above $\sim$ 10 TeV. The survival probability models with and without photon-ALP mixing seems to coincide. However, for better comparison between the survival probabilities including and excluding mixing in the source we introduce the photon survival probability ratio, $P_{ratio}$:

$$P_{ratio} = \frac{P_{\gamma\to\gamma}\,(\text{w. source})}{P_{\gamma\to\gamma}\,(\text{w/o. source})} \tag{6.14}$$

This ratio, plotted against energy, can be seen in Figure 6.3b.




(a) (b)

Figure 6.3: Comparison of photon survival probabilities for models with and without photon-ALP mixing in the source. (a) survival probabilities for both models including a model of EBL attenuation. (b) Ratio of survival probabilities of the two models.

It can be seen that the ratio $\approx 1$ across the entire energy range, indicating that the photon survival probability can be modeled without photon-ALP mixing in the source.
Therefore, further results are based on a photon-ALP beam propagating through three environments: the host galaxy, extragalactic space, and the Milky Way.

When modeling the observed spectrum one has to choose a model of the intrinsic spectrum. In this work we could either use a log-parabola or a power-law, which is a generalized model of the log-parabola. To see which of these models are preffered by the data we construct a best-fit models for both of them using the Franceschini et al. (2017) [35] EBL model. their resulting $\chi^2$-values can be seen in Table 6.2

| | Power-law | Log-parabola |
|---|---|---|
| $\chi^2$ | | |

Table 6.2: Caption

Comparison between the $\chi^2$-values show that the model of the observed spectrum including a power-law intrinsic spectrum is preferred by the LHAASO data. Therefore, we will adopt the a power-law model of the intrinsic spectrum in the further results.

### 6.3.2 Starburst galaxy

We construct the modeled SEDs of GRB221009A using Equation (6.2), where the photon survival probability entering the equation, includes both the contribution of the EBL attenuation model (hereafter the EBL-only model) and the photon-ALP oscillation model (hereafter the ALP model). Their photon survival probabilities are given by Equations (6.3) and (3.65). These probabilities are computed using the parameters listed in Table 6.1. In addition, both SED models adopt the four EBL models introduced in Section 4. In this subsection, we first consider the case of a starburst-type host galaxy.
Figure 6.4 shows the distribution of $\Delta\chi^2$-values across the ALP parameter space for the modeled SEDs using four different EBL models. Each pixel in the plot corresponds to a specific combination of ALP mass and photon-ALP coupling, with the color encoding corresponding to the $\Delta\chi^2$ value that originates from the hypothesis test, comparing the $\chi^2$ of the EBL-only model and the ALP model (see Section 6.2.1). For $\Delta\chi^2 > 0$ the null hypothesis is rejected, and the data favours the ALP model, while for $\Delta\chi^2 < 0$ the data prefers the EBL-only model.
The four panels of Figure 6.4, correspond to the SEDs constructed using the following EBL models: Franceschini et al. (2017) (top-left), Saldana-Lopez et al. (2021) (top right), the upper bound of Saldana-Lopez et al. (bottom-left), and Finke et al. (2010) (bottom-right). The color scale ranges from black ($\Delta\chi^2 < 0$) to bright yellow ($\Delta\chi^2 > 0$).
In all panels of Figure 6.4 a large section of the ALP parameter space, particularly where $g_{a\gamma\gamma} > 2\times10^{-11}$ GeV$^{-1}$ and $m_a < 100$ neV, the data favours the EBL-only model, suggesting that ALPs are not required to explain the LHAASO dataset for these ALP parameters. In contrast, some ALP parameter combinations yield positive $\Delta\chi^2$ values - especially for the SEDs using the Franceschini et al. (2017) and both Saldana-Lopez et al. (2021) EBL models. This indicates that in some regions the inclusion of ALPs improves the SED models fit to the data. The ALP parameters associated with the highest positive $\Delta\chi^2$ values, are referred to as the *best-fit* models.
From the upper right and lower left panels in Figure 6.4, it appears that the best-fit model is associated with $m_a = 16$ neV and $g_{a\gamma\gamma} = 10^{-10}$ GeV$^{-1}$. However, this coupling is excluded by constraints from CAST [6], which is why we rule-out this observation. As a consequence, for each of the panels in Figure 6.4, the best-fit model is associated with $m_a = 251$ neV and $g_{a\gamma\gamma} = 3.16 \times 10^{-11}$ GeV$^{-1}$. Table 6.3 summarizes the corresponding $\Delta\chi^2$ and $\chi^2$ values for the best-fit model. The corresponding SED models for the best-fit

| EBL model | $\Delta\chi^2$ | $\chi^2_{ALP}$ | $\chi^2_{EBL}$ |
|---|---|---|---|
| Franceschini et al. (2017) | 4.92 | 6.72 | 11.6 |
| Saldana-Lopez et al. (2021) | 3.83 | 11.4 | 15.2 |
| Saldana-Lopez et al. (2021) upper bound | 3.93 | 9.76 | 13.7 |
| Finke et al. (2022) | 1.57 | 8.20 | 9.77 |

Table 6.3: The $\Delta\chi^2$ and $\chi^2$ values for the model SEDs including photon-ALP mixing or only EBL attenuation based on four different EBL models.

ALP parameters are shown in Figure 6.5a, along with their residuals. Each curve corre-

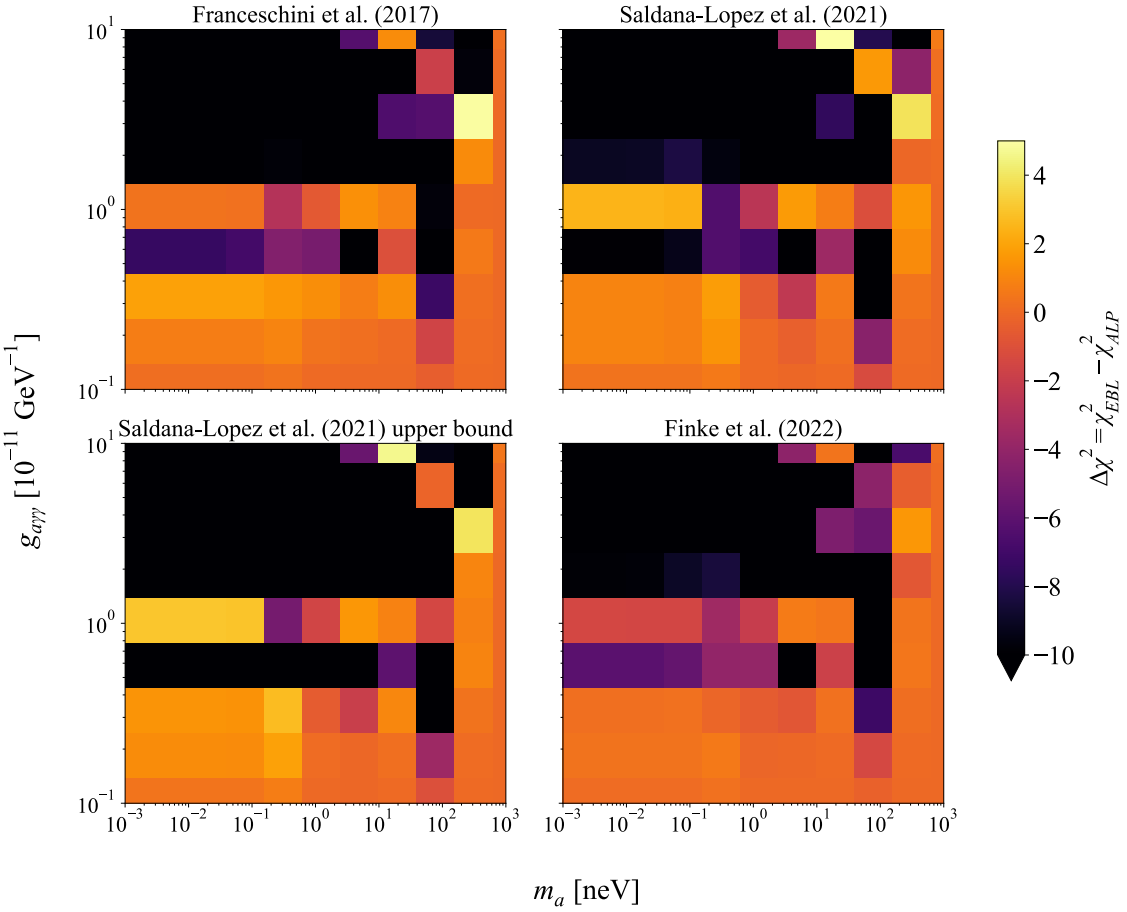
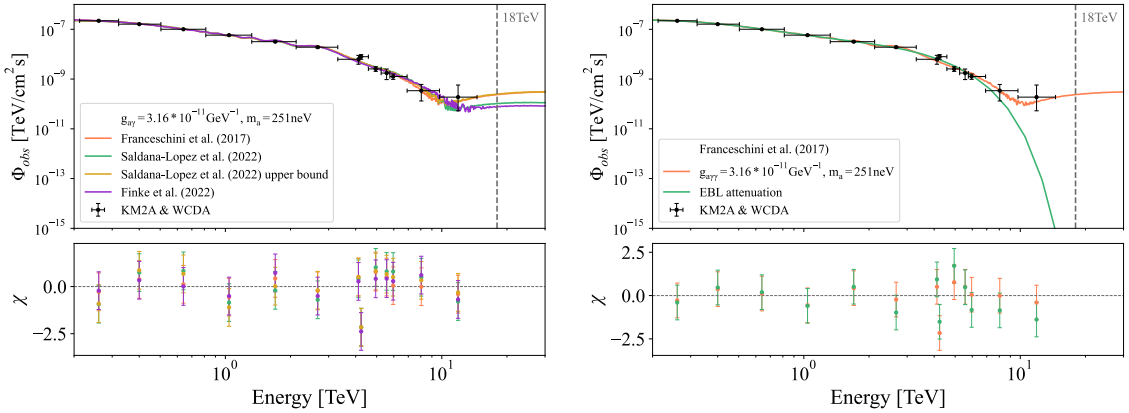



Figure 6.4: Distribution of $\Delta\chi^2$ values across the ALP parameter space. The SED models associated with each combination of ALP parameters are based on four different EBL models: Franceschini et al. (2017) (upper-left), Saldana-Lopez et al. (2021) (upper-right), the upper bound of Saldana-Lopez et al. (lower-left), and Finke et al. (2022) (lower-right). This plot effectively compares the EBL-only and ALP models through the $\Delta\chi^2$-value.

sponds to the different models from Figure 6.4.
Consider Figure 6.5a. Below $\sim$ 2 TeV, the SED models using Franceschini et al. (2017) and Finke et al. (2022) yield nearly identical residuals. At higher energies, the predictions from the SED model using Franceschini et al. (2017) and the upper-bound Saldana-Lopez et al. (2021) EBL models align well with the datapoints, and they contribute with very small residuals to the overall $\chi^2$. In particular, the SED constructed using Franceschini et al. (2017) has the lowest overall $\chi^2$ (see Table 6.3), deeming it the preferred model for this best-fit scenario.
In Figure 6.5b the best-fit ALP model and the associated EBL-only model are compared. As expected, the EBL-only model (green curve) decreases significantly at energies above 10 TeV, whereas the ALP model effectively enhances the contribution to the SED. Comparison between their residuals show a conformity at energies below 3 TeV, both laying within the $1\sigma$ uncertainties. However, at higher energies, the EBL-only model deviates more from the ALP model, with some residuals contributing with more than $1\sigma$ to the $\chi^2$. This deviation at VHE is expected due to absorption on the EBL. The ALP model maintain a better agreement with the high-energy region of the SED, remaining mostly within $1\sigma$ even around 13 TeV.




(a) (b)

Figure 6.5: Model SEDs based on the EBL models by Finke et al. (2022), Franceschini et al. (2017), and Saldana-Lopez et al. (2021) (both their best-fit and upper-bound) with the residuals. These are compared with the observed SED from LHAASO. (a) The four best-fit models and (b) comparison of the EBL-only and ALP models. Residuals are shown below the spectra, indicating the deviation of model predictions from the observed LHAASO data points. See Appendix A.2 for the fitting parameters.

### 6.3.3 Regular spiral galaxy

Adopting the terminology introduced in Section 6.3.2 and following the same analysis framework, we now consider a spiral-type host galaxy.

Figure 6.6 shows the distribution of $\Delta\chi^2$ values across the ALP parameter space. Across all four panels, a broad region of the ALP parameter space with $m_a < 100$ neV and $g_{a\gamma\gamma} > 10^{-11}$ GeV$^{-1}$ favours the EBL-only model, disfavouring the need for ALPs to explain the LHAASO data. The SED models based on the best-fit[1] and upper-bound Saldana-Lopez et al. (2021) EBL models include a few small regions where $\Delta\chi^2 > 0$. The best-fit ALP model corresponds to $m_a = 1.58 \times 10^{-2}$ neV and $g_{a\gamma\gamma} = 0.56 \times 10^{-11}$ GeV$^{-1}$, although the statistical significance remains low.

For the SED models based on Franceschini et al. (2017) and Finke et al. (2022), the entire parameter space yields low $\Delta\chi^2$ values. In particular, the model based on Finke et al. (2022) disfavours the ALP hypothesis almost entirely, with no region providing a significant improvement over the EBL-only model. The SED model using Franceschini et al. exhibits slightly higher $\Delta\chi^2$ in some regions, but none exceeds a $\Delta\chi^2$ of 2 (see Table 6.4). Therefore, these models do not provide statistically significant evidence for ALPs.

[1]This best-fit model should not be confused with the best-fit model of the SEDs.

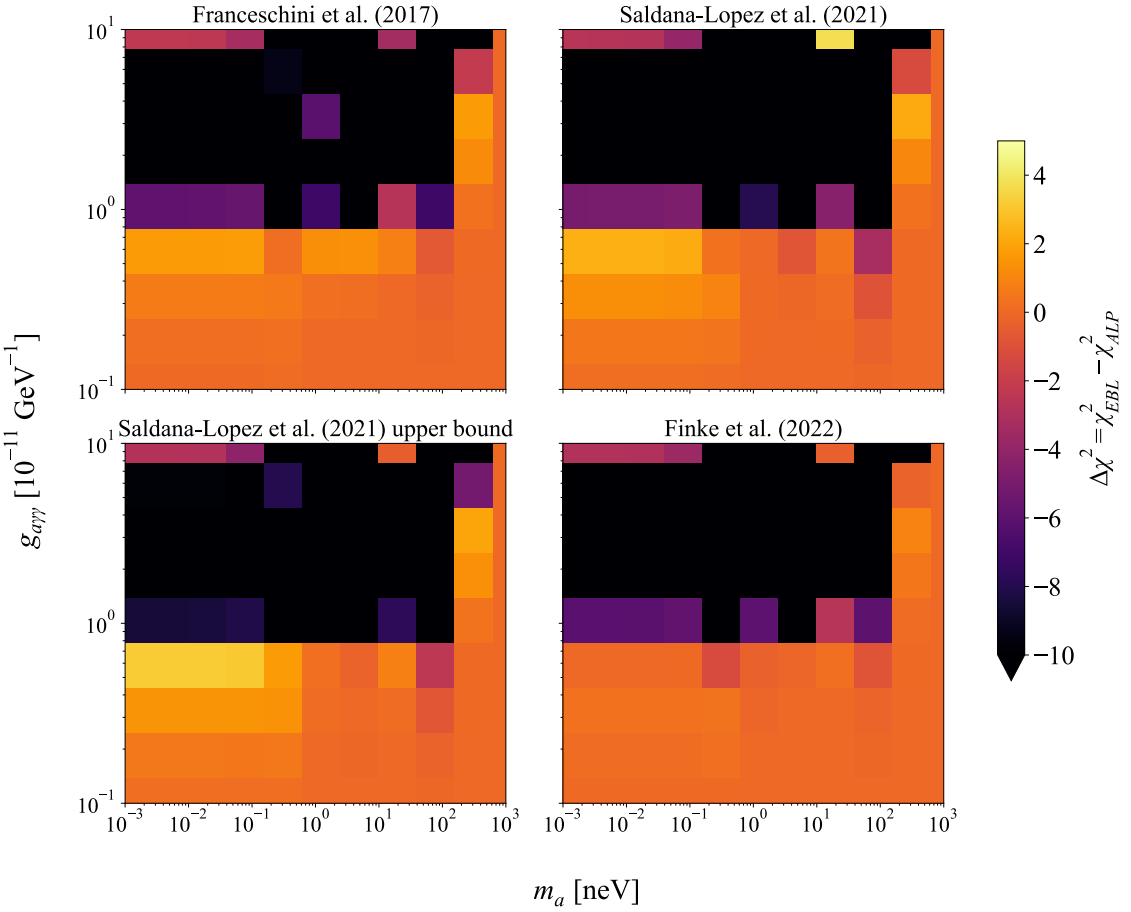


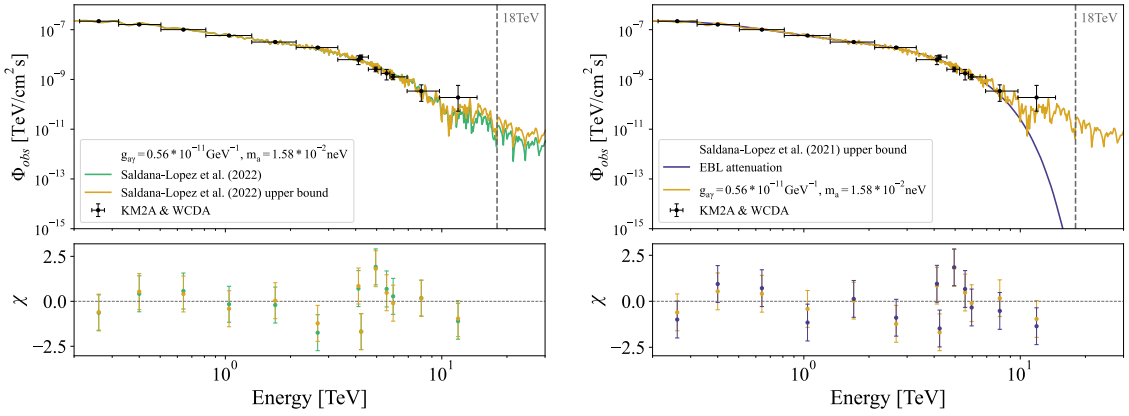

Figure 6.6: Model SEDs based on the EBL models by Finke et al. (2022), Franceschini et al. (2017), and Saldana-Lopez et al. (2021) (both their best-fit and upper-bound) with the residuals. These are compared with the observed SED from LHAASO. (a) The four best-fit models and (b) comparison of the EBL-only and ALP models. Residuals are shown below the spectra, indicating the deviation of model predictions from the observed LHAASO data points. See Appendix A.2 for the fitting parameters.

| EBL model | $\Delta\chi^2$ | $\chi^2_{ALP}$ | $\chi^2_{EBL}$ |
|---|---|---|---|
| Franceschini et al. (2017) | 1.73 | 9.91 | 11.6 |
| Saldana-Lopez et al. (2021) | 2.35 | 12.8 | 15.2 |
| Saldana-Lopez et al. (2021) upper bound | 3.17 | 10.5 | 13.7 |
| Finke et al. (2022) | 0.92 | 9.79 | 9.77 |

Table 6.4: The $\Delta\chi^2$ and $\chi^2$ values comparing the ALP and EBL-only models for four different EBL templates, assuming a spiral-type host galaxy

The SED models based on both Saldana-Lopez et al (2022) models for the best-fit ALP parameters are shown in Figure 6.7a. The residuals of both models largely coincide across the entire energy range, except for small deviations around 3–4 TeV. The SED model based on the upper-bound of Saldana-Lopez et al. (2021) has the smallest $\chi^2$ value of the two models in Figure 6.7a (see Table 6.4), implying that this model is favoured by the data.

Figure 6.7b shows the best-fit ALP model based on the upper-bound of Saldana-Lopez et al. (2021) and the associated EBL-only model. At low energies, the ALP model exhibits a strong oscillatory behaviour resulting in a deviation in the residuals from the EBL model. At energies above 7 TeV, the residuals of the ALP and EBL-only models diverge, while in the intermediate energy range, they remain similar.




(a) (b)

Figure 6.7: Model SEDs based on the best-fit and upper-bound of the Saldana-Lopez et al. (2021) EBL model with the residuals. These are compared with the observed SED from LHAASO. (a) The two best-fit models and (b) comparison of the EBL-only and ALP models. Residuals are shown below the spectra, indicating the deviation of model predictions from the observed LHAASO data points. See Appendix A.2 for the fitting parameters.

# Chapter 7

# Discussion

In this thesis, data measured by LHAASO were analysed using different SED models to determine whether ALPs are needed to explain the VHE photons observed from GRB221009A. The SED was modeled either with standard EBL attenuation or by including photon-ALP mixing. Since both approaches depend on the choice of EBL model, several models were tested. The models that were not excluded by the data: Franceschini et al. (2017) [35], Finke et al. (2022) [32], and Saldana-Lopez et al. (2021) [79] (both their best-fit and upper bound) were used throughout the analysis.

To incorporate both EBL attenuation and photon-ALP oscillations into the SED model, the photon survival probability was calculated using the `GammaALPs` package [66]. Photon-ALP mixing within the GRB source was found to be negligible, consistent with the results of Galanti et al. [38]. The subsequent analysis therefore focused on mixing effects during propagation through the host galaxy, extragalactic space, and the Milky Way. The magnetic fields in these environments were modeled following Galanti et al. [38], assuming two host galaxy scenarios: a starburst and a spiral galaxy.

Across both the starburst and spiral galaxy scenarios, the results consistently show that the EBL-only model provides an adequate fit to the LHAASO data, with no statistically significant improvement from including photon-ALP mixing. In particular, large regions of the ALP parameter space (for $g_{a\gamma\gamma} > 10^{-11}$ GeV$^{-1}$ and $m_a < 100$ neV) are disfavored by the data, suggesting that ALPs are not required to explain the observed SED of GRB221009A within the tested parameter space. Some limited regions of the ALP parameter space did show small improvements in the model fit, depending on the host galaxy scenario. For the starburst host galaxy, using the Franceschini et al. (2017) EBL model, the best-fit parameters were $m_a = 251$ neV and $g_{a\gamma\gamma} = 3.16 \times 10^{-11}$ GeV$^{-1}$, with a corresponding $\Delta\chi^2 = 4.92$. For the spiral galaxy, the best-fit parameters were $m_a = 1.58 \times 10^{-2}$ neV and $g_{a\gamma\gamma} = 0.56 \times 10^{-11}$ GeV$^{-1}$ with $\Delta\chi^2 = 3.17$. However, these improvements ($\Delta\chi^2 \lesssim 5$) are not statistically significant and do not provide strong evidence in favour of ALPs.

When comparing the best-fit ALP parameters from both starburst and spiral scenarios to current exclusion limits (see Figure **??**), it becomes clear that these regions are strongly constrained by astrophysical observations. However, as discussed in Section 3.3, these limits depend on specific model assumptions (e.g., magnetic field strengths, ALP production mechanisms) and should be interpreted with caution. Importantly, the ALP parameter values obtained in this analysis are not excluded by current laboratory experiments such as CAST [6] and OSQAR [9], leaving open the possibility for future experimental exploration.

Lastly, it is worth noting that the magnetic field strengths adopted for the host galaxies in this analysis were the lower limits presented by Galanti et al. [38]. Exploring the impact

of stronger magnetic fields—up to $B = 50$ $\mu$G for starburst galaxies and $B = 20$ $\mu$G for spiral galaxies—could potentially lead to better fits. Additionally, the entire analysis relied on LHAASO data reconstructed under the assumption of a power-law with exponential cutoff (PLEC) spectrum. In future work, it would be valuable to repeat this analysis using data reconstructed with a log-parabola model, which extends to photon energies up to $\sim 18$ TeV. Such efforts may provide further insights into the nature of VHE emission from GRBs and the role of ALPs in high-energy astrophysics.

# Chapter 8

# Summary + outlook

In this thesis, we investigated whether ALPs could explain the VHE photons observed by LHAASO from GRB221009A. We constructed models of the observed SED based on an intrinsic log-parabola shape and photon survival probabilities that account for attenuation EBL, with and without the addition of photon–ALP oscillations.
The analysis employed a maximum likelihood approach over a broad ALP parameter space ($m_a \sim 10^{-3}$–$10^{3}$ neV, $g_{a\gamma\gamma} \sim 10^{-12}$–$10^{-10}$ GeV$^{-1}$), using four different EBL models (Franceschini et al. (2017), Finke et al. (2022), and the best-fit and upper-bound models from Saldana-Lopez et al. (2021)). Photon propagation was modeled through the host galaxy (assuming both starburst and spiral morphologies), extragalactic space, and the Milky Way, with magnetic fields in each environment incorporated following established models.
Our results show that for both host galaxy scenarios, a large region of the ALP parameter space, particularly for $m_a < 100$ neV and $g_{a\gamma\gamma} > 10^{-11}$ GeV$^{-1}$, favours a model based on EBL attenuation. This suggests that ALPs are not required to explain the VHE gamma rays from GRB221009A within the tested parameter space. A modest improvement in the fit was found for the starburst galaxy scenario at $m_a = 251$ neV and $g_{a\gamma\gamma} = 3.16 \times 10^{-11}$ GeV$^{-1}$, though this result has low statistical significance and lies in tension with some astrophysical constraints, while remaining compatible with current experimental bounds from CAST.
In summary, our analysis indicates that photon–ALP mixing is not necessary to explain the LHAASO observations of GRB221009A.
There are several avenues for improving and extending this analysis in future work. First, the present study relied on LHAASO data reconstructed using a power-law with exponential cutoff (PLEC) model, with photon energies extending up to 13 TeV. An alternative LHAASO dataset, reconstructed with a log-parabola model, reaches photon energies up to 18 TeV and could provide additional sensitivity to ALP effects at the highest energies. Second, the magnetic field model for the host galaxy could be refined. This work used generic models for spiral and starburst galaxies. A more targeted approach would involve adopting Galactic Magnetic Field models specific to the Milky Way, as has been done in related studies [16]. Such modeling would allow a more direct comparison between different works and help quantify the influence of host galaxy assumptions on ALP constraints. Finally, one could adopt another hypothesis-testing framework, used to evaluate the statistical preference for or against specific region of the ALP parameter space.

# Appendix A

# Appendix

## A.1 Photon-ALP interactions

### A.1.1 Derivation of the photon-ALP beam equation

The photon-ALP beam equation can be derived from the Euler-Lagrange equation. For an ALP field Equation (3.24) reads:

$$\partial_\mu \left( \frac{\partial \mathcal{L}}{\partial \left( \partial_\mu a \right)} \right) - \frac{\partial \mathcal{L}}{\partial a} = 0 \tag{A.1}$$

Only the terms in the Lagrangian density (Equation (3.20)) that include the ALP field, $a$, or its derivative, $\partial_\mu a$ contribute to the above Euler-Lagrange equation. The two terms in the above equation are computed separately for the sake of clarity.

$$\partial_\mu \left[ \frac{1}{2} \frac{\partial}{\partial \left( \partial_\mu a \right)} \left( \partial_\alpha a \partial^\alpha a - m_a^2 a^2 \right) \right] = \frac{1}{2} \partial_\mu \left[ \frac{\partial}{\partial \left( \partial_\mu a \right)} \left( \partial_\alpha a \partial_\beta g^{\beta\alpha} a \right) \right] \tag{A.2}$$

$$= \frac{1}{2} \partial_\mu \left( g^{\beta\alpha} \left[ \delta^\mu_\alpha \partial_\beta a + \partial_\alpha a \delta^\mu_\beta \right] \right) \tag{A.3}$$

$$= \partial_\mu \partial^\mu a \tag{A.4}$$

$$\frac{\partial}{\partial a} \left( \frac{1}{2} \left[ \partial_\alpha a \partial^\alpha a - m_a^2 a^2 \right] + \frac{g_{a\gamma\gamma}}{4} F_{\mu\nu} \tilde{F}^{\mu\nu} a \right) = -m_a^2 a + \frac{g_{a\gamma\gamma}}{4} F_{\mu\nu} \tilde{F}^{\mu\nu} \tag{A.5}$$

$$= -m_a^2 a + \frac{g_{a\gamma\gamma}}{4} \mathbf{E} \cdot \mathbf{B} \tag{A.6}$$

Where $g_{\beta\alpha} = g^{\beta\alpha}$ is the metric tensor. The combined Euler-Lagrange equation for the ALP field reads

$$\partial_\mu \partial^\mu a + m_a^2 a - \frac{g_{a\gamma\gamma}}{4} \mathbf{E} \cdot \mathbf{B} = 0 \tag{A.7}$$

The procedure is the same four the $A_\nu$ fields. Here, Equation (3.24) becomes

$$\partial_\mu \left( \frac{\partial \mathcal{L}}{\partial \left( \partial_\mu A_\nu \right)} \right) - \frac{\partial \mathcal{L}}{\partial A_\nu} = 0 \tag{A.8}$$

The second term of the Euler-Lagrange equation is 0 because the Lagrangian density does not possess a direct dependence of $A_\mu$. Only derivative terms such as $\partial_\mu A_\nu$ contribute to the Euler-Lagrange equation. These are found in terms that include the field strength tensor and/or its dual. The calculation is initiated with a calculation of the derivative of each of these terms:

$$\frac{\partial}{\partial\left(\partial_\mu A_\nu\right)}\left(-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}\right) = -\frac{1}{4}\left[\frac{\partial F_{\alpha\beta}}{\partial\left(\partial_\mu A_\nu\right)}F^{\alpha\beta} + F_{\alpha\beta}\frac{\partial F^{\alpha\beta}}{\partial\left(\partial_\mu A_\nu\right)}\right] \tag{A.9}$$

$$= -\frac{1}{4}\left[\frac{\partial\left(\partial_\alpha A_\beta - \partial_\beta A_\alpha\right)}{\partial\left(\partial_\mu A_\nu\right)}F^{\alpha\beta} + F_{\alpha\beta}g^{\alpha\rho}g^{\sigma\beta}\frac{\partial\left(\partial_\rho A_\sigma - \partial_\sigma A_\rho\right)}{\partial\left(\partial_\mu A_\nu\right)}\right] \tag{A.10}$$

$$= -\frac{1}{4}\left[\left(\delta^\mu_\alpha\delta^\nu_\beta - \delta^\mu_\beta\delta^\nu_\alpha\right)F^{\alpha\beta} + F^{\rho\sigma}\left(\delta^\mu_\rho\delta^\nu_\sigma - \delta^\mu_\sigma\delta^\nu_\rho\right)\right] \tag{A.11}$$

$$= -F^{\mu\nu} \tag{A.12}$$

$$\frac{\partial}{\partial\left(\partial_\mu A_\nu\right)}\left(\frac{g_{a\gamma\gamma}}{4}F_{\alpha\beta}\tilde{F}^{\alpha\beta}a\right) = \frac{g_{a\gamma\gamma}}{4}\left[\frac{\partial F_{\alpha\beta}}{\partial\left(\partial_\mu A_\nu\right)}\tilde{F}^{\alpha\beta} + F_{\alpha\beta}\frac{\partial}{\partial\left(\partial_\mu A_\nu\right)}\left(\frac{1}{2}\epsilon^{\alpha\beta\rho\sigma}F_{\rho\sigma}\right)\right]a \tag{A.13}$$

$$= \frac{g_{a\gamma\gamma}}{4}\left[\frac{\partial\left(\partial_\alpha A_\beta - \partial_\beta A_\alpha\right)}{\partial\left(\partial_\mu A_\nu\right)}\tilde{F}^{\alpha\beta} + \frac{1}{2}F_{\alpha\beta}\epsilon^{\alpha\beta\rho\sigma}\frac{\partial\left(\partial_\rho A_\sigma - \partial_\sigma A_\rho\right)}{\partial\left(\partial_\mu A_\nu\right)}\right]a \tag{A.14}$$

$$= \frac{g_{a\gamma\gamma}}{4}\left[\left(\delta^\mu_\alpha\delta^\nu_\beta - \delta^\mu_\beta\delta^\nu_\alpha\right)\tilde{F}^{\alpha\beta} + \tilde{F}^{\rho\sigma}\left(\delta^\mu_\rho\delta^\nu_\sigma - \delta^\mu_\sigma\delta^\nu_\rho\right)\right]a \tag{A.15}$$

$$= g_{a\gamma\gamma}\tilde{F}^{\mu\nu}a \tag{A.16}$$

I have used the antisymmetric property of the field strength tensor i.e. $F^{\mu\nu} = -F^{\nu\mu}$, its definition in terms of a vector potential $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and the definition of its dual $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$. The Euler-Lagrangian for the $A_\nu$ fields become

$$\partial_\mu\left[\frac{\partial}{\partial\left(\partial_\mu A_\nu\right)}\left(-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{g_{a\gamma\gamma}}{4}F_{\mu\nu}\tilde{F}^{\mu\nu}a\right)\right] = -\partial_\mu F^{\mu\nu} + g_{a\gamma\gamma}\partial_\mu\tilde{F}^{\mu\nu}a \tag{A.17}$$

$$-\partial_\mu F^{\mu\nu} + g_{a\gamma\gamma}\partial_\mu\tilde{F}^{\mu\nu}a = 0 \tag{A.18}$$

## A.2 Fitting parameters

In the following Table the parameter $\phi_0$ is given in units of $10^{-8}\mathrm{cm}^{-2}\mathrm{s}^{-1}\mathrm{TeV}^{-1}$.

Fitting parameters associated with the modeled spectra using the best-fit ALP parameters $m_a = 251$ neV and $g_{a\gamma\gamma} = 3.16 \times 10^{-11}$ GeV$^{-1}$.

| EBL model | $\phi_0$ | $\alpha$ | $\beta$ |
|---|---|---|---|
| Franceschini et al. (2017) | 35.53 | 2.07 | 0.116 |
| Saldana-Lopez et al. (2021) | 29.02 | 2.19 | 0.149 |
| Saldana-Lopez et al. (2021) upper bound | 37.09 | 2.02 | 0.131 |
| Finke et al. (2022) | 25.95 | 2.28 | 0.139 |

Table A.1: The fitting parameters $\phi_0$, $\alpha$, and $\beta$ for modeled spectra including photon-ALP mixing or only EBL attenuation based on four different EBL models and assuming a starburst-type host galaxy.

Fitting parameters associated with the modeled spectra using the follwing best-fit ALP parameters in order: $m_a = 6.31 \times 10^{-2}$ neV and $g_{a\gamma\gamma} = 0.56 \times 10^{-11}$ GeV$^{-1}$, $m_a = 1.58 \times 10^{-2}$ neV and $g_{a\gamma\gamma} = 0.56 \times 10^{-11}$ GeV$^{-1}$ ,$m_a = 251$ neV and $g_{a\gamma\gamma} = 3.16 \times 10^{-11}$ GeV$^{-1}$.

| EBL model | $\phi_0$ | $\alpha$ | $\beta$ |
|---|---|---|---|
| Franceschini et al. (2017) | 38.10 | 2.18 | 0.172 |
| Saldana-Lopez et al. (2021) | 31.19 | 2.30 | 0.207 |
| Saldana-Lopez et al. (2021) upper bound | 39.78 | 2.14 | 0.189 |
| Finke et al. (2022) | 24.88 | 2.34 | 0.164 |

Table A.2: The fitting parameters $\phi_0$, $\alpha$, and $\beta$ for modeled spectra including photon-ALP mixing or only EBL attenuation based on four different EBL models and assuming a spiral-type host galaxy.

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